

# Longwall mining

R M BHATTACHARJEE

**Department of Mining Engineering**

**IIT(ISM), Dhanbad**

**Lecture Plan**

**B.Tech. (ME) & Dual Degree**

**Session: 2018-19**

**Semester - Monsoon**

**Subject: Underground Coal Mining**

**Subject Code: MEC 15103)**

**LTP: 3-0-0**

**Longwall Mining – 1-0-0**

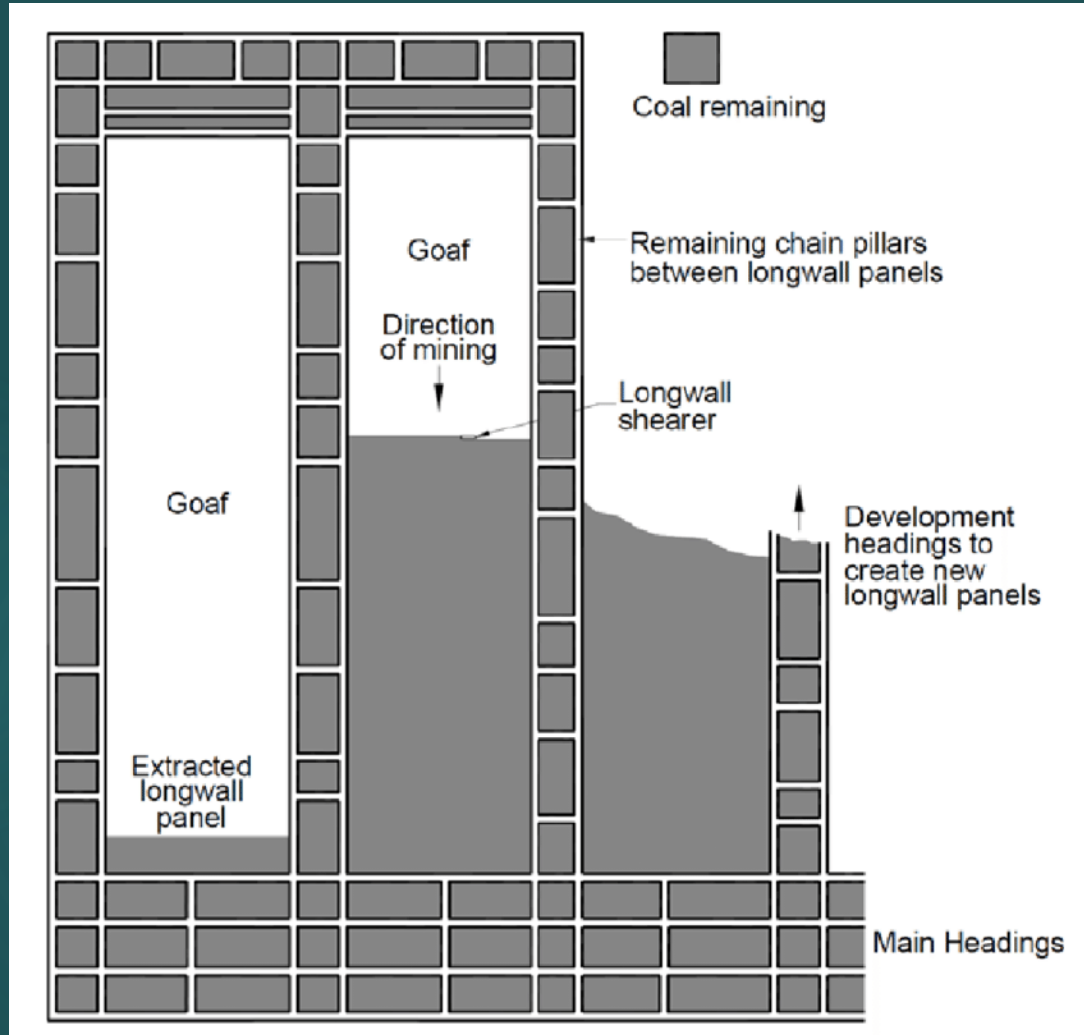
|           |                                                                                                                                                                                                |          |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>1.</b> | <b>Factors affecting longwall mining; Longwall mine layout; Advancing and retreating faces, Single versus double unit longwall faces, Orientation of longwall faces, Longwall face layouts</b> | <b>4</b> |
| <b>1.</b> | Longwall panel development, Factors affecting length and width of longwall panel, Drivage equipment, Support, Ventilation , Single versus multiple heading gate roads                          | 2        |
| <b>1.</b> | Extraction of Longwall panel, Working with shearer and plough; Longwall face equipment                                                                                                         | 2        |
| <b>1.</b> | Strata control in and around longwall panels; Support system of longwall face and gate roads , Monolithic packing in longwall advancing gate roads                                             | 4        |
| <b>1.</b> | Local and main fall, indications of roof weighting, measures to bring down roof at regular interval; air blast and measures to minimize its effects                                            | 2        |

# Longwall mining

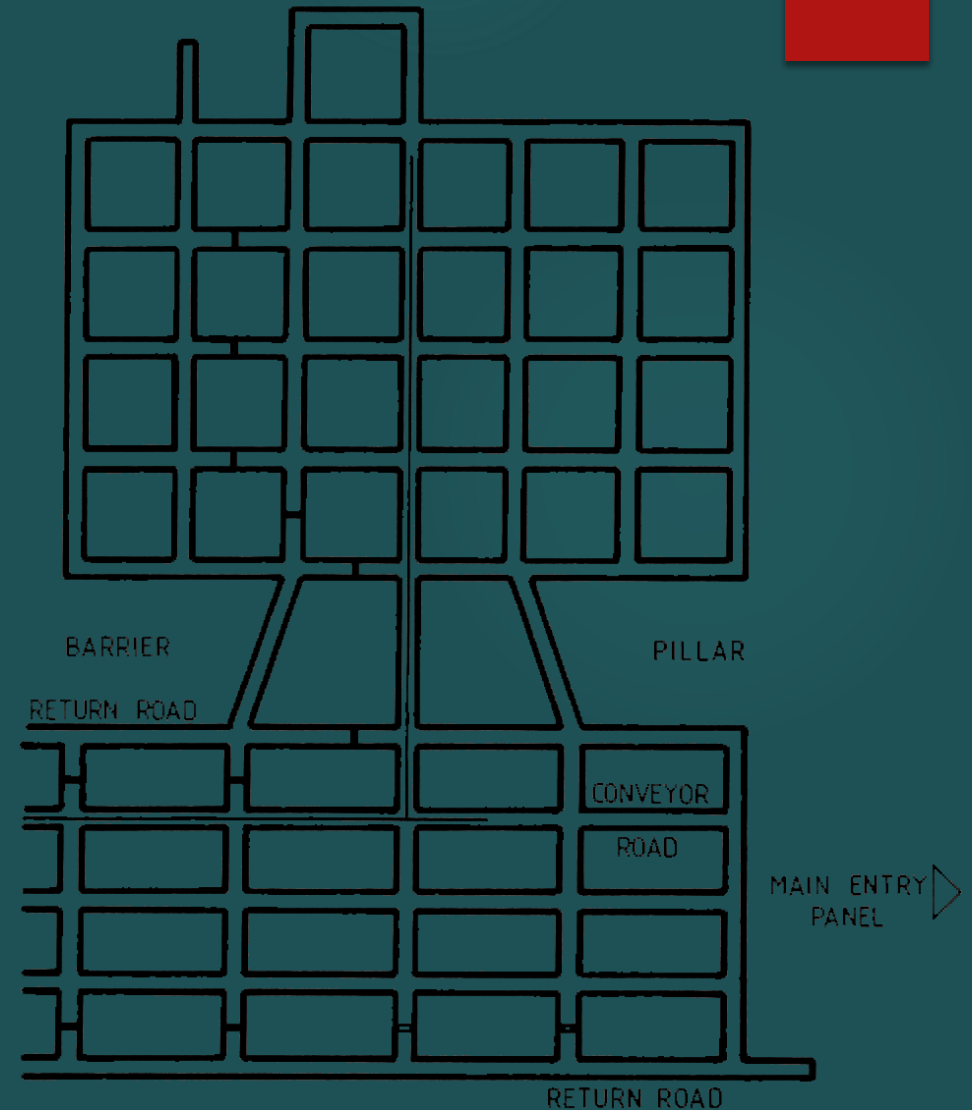
Difference between longwall and bord & pillar

- ▶ Longer pillar compared to bord & Pillar mining
- ▶ Larger coal block
- ▶ Scope of high degree of mechanization
- ▶ Straight goaf edge

# Longwall and Bord and Pillar panels

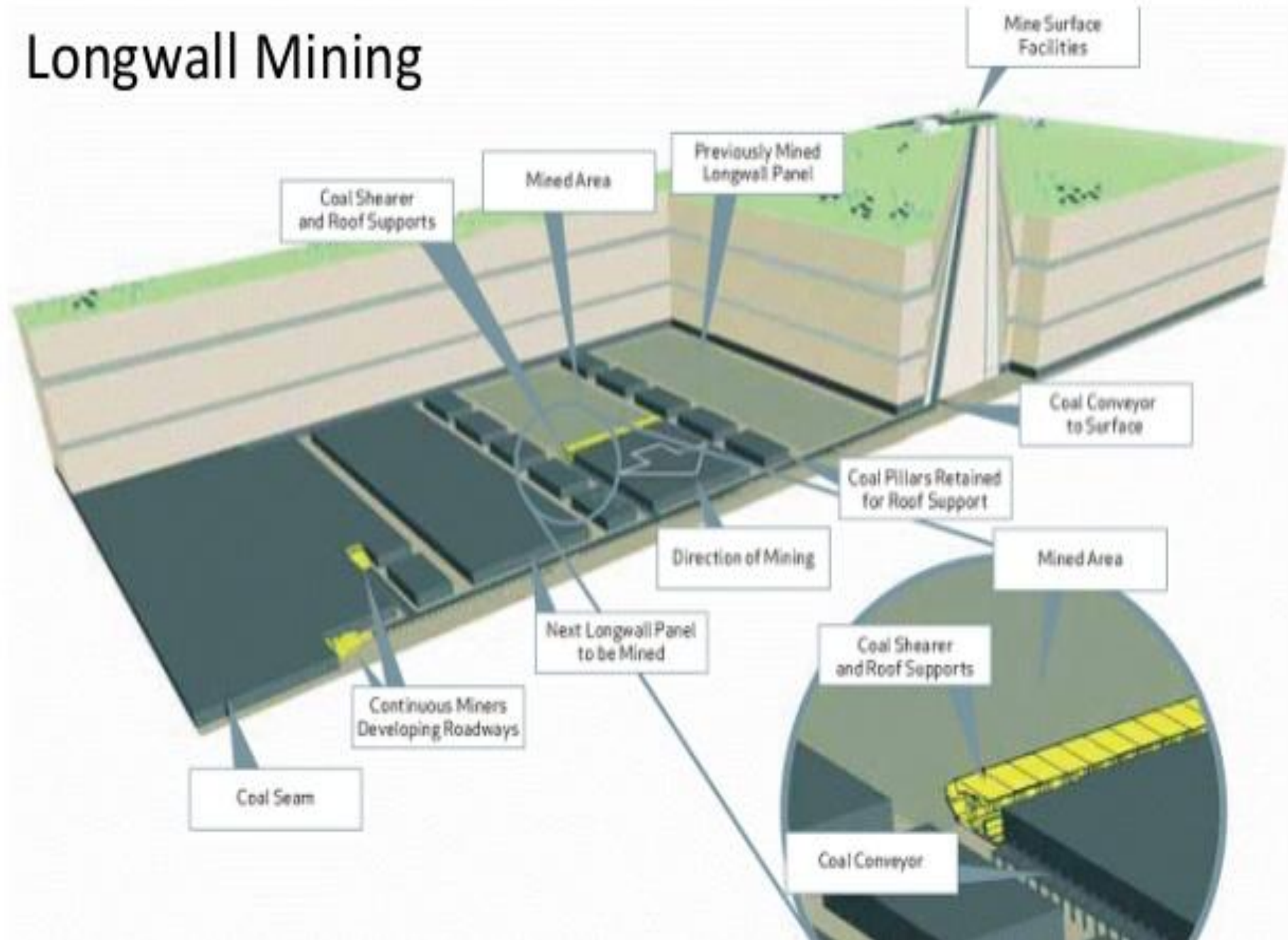


Longwall Layout



Bord and Pillar Layout

# Longwall Mining



## Global Longwall Production: When India Ended its Experiment

|             |   | No. of Faces | Best<br>(mtpa) | Average<br>(mtpa) |
|-------------|---|--------------|----------------|-------------------|
| • USA       | : | 49           | 7.2            | 4.3               |
| • Australia | : | 24           | 5.7            | 2.8               |
| • China     | : | 600          | 8.9            | 0.8               |
| • Russia    | : | 120          | 4.1            | 0.5               |
| • Ukraine   | : | 170          | 0.5            | 0.2               |
| • India     | : | 02           | 2.1            | 0.1               |

# The CIL Experience

| Mine             | Year       | Source                | Supports (Te)  | Capacity(TPD) | Remarks                                                                                                                        |
|------------------|------------|-----------------------|----------------|---------------|--------------------------------------------------------------------------------------------------------------------------------|
| Moonidih         |            |                       |                |               |                                                                                                                                |
| i) ML-IV         | 1978       | Polish, Meco, Gullick | 4 X 380        | 860           | -Mismatch with geominig condition<br>-Customs clearance , testing of equipment ,sources of spares led to cost over-run project |
| ii) ML-V         | 1979       |                       | 4X 420         | 840           |                                                                                                                                |
| iii) ML-VI       | 1981       |                       | 4 X 280        | 470           |                                                                                                                                |
| iv) ML-VII       | 1982       |                       | 4 X 870        | 3200          |                                                                                                                                |
| Seetalpur        | 1982       | GD/UK                 |                | 800           | - In compatibility with geominig condition                                                                                     |
| Dhemo Main       | 1987       | GD/UK                 | 4 X 480        | 550           | Acquisition of land coupled with sourcing spares frm overseas – led to cost over run .                                         |
| Jhanjra (1 &2)   | 1982       | USSR                  |                | 1476          | Acquisition of land coupled with sourcing spares frm overseas – led to cost over run                                           |
| Kottadih         | 1997       | CdF, France           |                |               | Moderately successful in 1 <sup>st</sup> panel – but mine collapsed                                                            |
| Pathakhera       | 1996       | MECO/UK               |                | 550           |                                                                                                                                |
| Churcha West     | 1989       | GD/UK                 |                |               | Collapsed                                                                                                                      |
| Jhanjra( 1 & 2)  | 1984       | USSR                  |                | 3000          |                                                                                                                                |
| Rajendra         |            | Chinese               |                | 2300          |                                                                                                                                |
| Jhanjra-MIC      | 2015       | CODCO-Chinese         |                |               | Under Installation                                                                                                             |
| Moonidih XVI (T) | 2015       | INDU-ZMJ              |                |               | 0.7 MTY Proj- Performance trail made and commercial production started .                                                       |
| Moonidih XV      | 2016 (Exp) | BUCYRUS               | 2 x 800 Shield |               | 2 MTY Proj- Incline drivage started and shaft sinking to be done                                                               |



# The SCCL Experience

| Mine       | Year | Source     | Supports (Te) | Capacity | Remarks          |
|------------|------|------------|---------------|----------|------------------|
| GDK 7 & 9  | 1983 | GD, UK     | 4 x 325 / 360 |          |                  |
| VK 7       | 1986 | GD, UK     | --- do ---    |          |                  |
| JK 5       | 1990 | GD, UK     | --- do ---    | 2000     | 1,000 HP Shearer |
| GDK 11 - 1 | 1991 | GD, UK     | 4 x 450       | 1600     |                  |
| GDK 11 - 2 | 1991 | Meco, UK   | 4 x 450       | 1600     |                  |
| GDK 11 - 3 | 1992 | GD, UK     | 4 x 450       | 1600     |                  |
| GDK 10A    | 1994 | Meco, UK   | 4 / 800       | 2000     |                  |
| PVK - 1    | 1995 | CME, China | 4 / 760       | 2200     |                  |
| PVK - 2    | 1996 | CME, China | 4 / 760       | 2200     |                  |

GD: Gullick Dobson; Meco: Dowty Meco; Capacity in TPD



# Reasons for Longwall Failure in India

- Installation in unsuitable conditions
- Lack of supplier accountability
- Unavailability of spare parts
- Short term supplier support
- Lack of training
- Lack of repair/rebuild facilities
- Poor mine infrastructure
- Lack of Management expertise



**Inability to  
Listen, Learn &  
Change  
Mindset**

# Advantages of longwall over bord & pillar:

- ▶ % extraction very high
- ▶ Continuous mining
- ▶ No fleeting of machinery, machine installed onetime and moves with the face
- ▶ Subsidence control is simple
- ▶ Strata control is effective with self-advancing powered support system
- ▶ Better ventilation
- ▶ Less risk of spontaneous heating
- ▶ High productivity
- ▶ Suitable for mass production
- ▶ Large scope of mechanization
- ▶ Concentration of activities – miners and machines

# where longwall mining may not be the best option

- ▶ Where market for the product is limited and not continuous; longwalls are not generally successful with intermittent operation.
- ▶ Where the mining lease contains a large number of seam discontinuities (longwall systems generally do not handle discontinuities of any significant size well) or other restrictions on the size of longwall blocks such as lease boundaries or existing workings;
- ▶ Where surface or sub-surface structures or landforms prohibit subsidence and restrict mining to first workings only
- ▶ Where capital required to invest in a longwall is problematic, at least until some income is available;
- ▶ Where infrastructure available is inadequate to handle high tonnages of coal or to provide adequate supplies; or
- ▶ Where labour force available will be difficult to train to the standard required for successful longwall operations (continuous miner operations tend to be somewhat less demanding)

# Requirements and constraints for high production longwalls

## ▶ Favourable Geomining Conditions

- ▶ Flat seams – not steeply inclined
- ▶ Medium to thick seam - not very thick seam
- ▶ Uniform thickness
- ▶ Easily cavable roof
- ▶ Not very high gas content – highly gassy seams need to be degasified prior to mining
- ▶ Free from geological discontinuities / anomalies
- ▶ Large property – size & reserve
- ▶ Free from important surface features like township, railways, rivers / waterbodies, etc.

# Requirements and constraints for high production longwalls

- ▶ Excellent mine plan and equipment layout allowing rapid development
- ▶ Minimum abutment pressure interaction between adjacent panels
- ▶ State of the art equipment –
  - ▶ Heavy duty, reliable,
  - ▶ Guaranteed supply of spares,
  - ▶ Standardization of equipment
  - ▶ Indigenisation of equipment and parts / components, spares

# Requirements and constraints for high production longwalls

- ▶ Efficient method of face move – less idle time – weekend transfer
- ▶ Compatible out-bye capabilities – transportation
- ▶ Capacity of coal transportation
- ▶ Quick and safe face installation
- ▶ Suitable organization structure
- ▶ Skilled workforce – multi-job workers
- ▶ Production and safety incentives

# Panel Layout

For successful operation of longwall mining, optimum panel layout should be designed with following issues for consideration:

- ▶ Length of Panel
- ▶ Width of panel
- ▶ Height of extraction
- ▶ Orientation of panel
- ▶ Cutting horizon
- ▶ Number of gate roads or entries into the panel
- ▶ Size of gate roads



# PANEL WIDTH OBJECTIVE

Optimisation required to maximise project value through:

- ▶ Maximising resource recovery –
  - ▶ Width of face and loss in barrier pillar
  - ▶ Orientation of panel – with respect to surface land constraints like rail, river, etc.; shape of property, regularity of lease boundary
- ▶ Maximising productivity –
  - ▶ cycle time vs face width
  - ▶ Maximum time in cutting cycle is sumping
- ▶ Minimum inter-panel interaction of abutment pressure

# PANEL WIDTH OBJECTIVE

- ▶ Minimising cost –
  - ▶ Development cost reduction with increase in panel width
  - ▶ Increase in capital cost due to increase in requirement of face machines
- ▶ Meet environmental requirements
  - ▶ Gas content
  - ▶ Roof condition
  - ▶ Dust
  - ▶ Heat
- ▶ Controlling risk
  - ▶ Minimum inter panel interaction of abutment pressure

The benefits of panel widening can only however, be realised up to the technical limits of the resource

# DEVELOPMENT REDUCTION

- ▶ Development is an inherent part of longwall mining
- ▶ The longer the block, the lower the total development metres required per longwall tonne made available
- ▶ When cutting, slowest part of each cutting cycle (shear) is at the face ends, so wider the face, greater the amount of coal available to cut at high rate
- ▶ A large proportion of strata control problems, or work required to avoid such problems, occurs at the face ends in the gate roads. Wider the face, fewer the number of gate roads within the longwall area and less the cost per tonne for gate road maintenance

# DEVELOPMENT REDUCTION

- ▶ The elimination of development metres from the plan represents a fixed reduction in underground activity
- ▶ Reduction in capital cost for development equipment. Much of the gate road equipment is the same regardless of the face width, or at least the extra cost for gate road equipment for the extra width is relatively small. The wider the block the lower the capital cost per longwall tonne
- ▶ Beyond a certain block width, the chain pillar size required for stability is not affected. The wider the face, the less coal is sterilised in these pillars

# Factors for deciding PANEL LENGTH

- ▶ Frequency of face transfer

- ▶ Productivity

- ▶ Cost

Relocating a longwall face from one block to another is a major logistical exercise involving high costs and during which there is no production (and hence no income) – the fewer moves required the better and the longer the blocks the fewer moves are required

- ▶ Loss of coal in end panel barrier

# Factors for deciding PANEL LENGTH

- ▶ Reliability of face equipment
- ▶ Reliability of out-bye equipment
- ▶ Transport of man, material & machine
- ▶ Time for development
- ▶ Geological discontinuity / anomaly
- ▶ Capability of gate road development
- ▶ Rate of face retreat

# Orientation of panel

- ▶ Orientation of panel – with respect to surface land constraints like rail, river, etc.; shape of property, regularity of lease boundary
- ▶ Magnitude and direction of horizontal in-situ stress
  - ▶ Major stress should not be perpendicular to face or side wall
- ▶ Seam gradient
  - ▶ Normally longwall face is laid along dip-rise and longwall face moves along strike direction
  - ▶ If the seam is too steep, stability of the machines and creep will be critical. It may require to orient the face in apparent dip direction to minimise the effect of seam inclination
  - ▶ Magnitude of cross-dip along strike galleries and stability of machine while development



## Cutting Horizon

- ▶ Depends on seam thickness and range of shearer cutting
- ▶ Normally face moves along floor to maintain horizon
- ▶ Better to cut the entire thickness, if possible, following floor for ease of maintaining horizon
- ▶ If floor is soft, may have to leave some coal against the floor.
- ▶ If seam thickness is more than cutting height, coal will be left either in floor or roof. If coal is prone to heating, leaving coal in roof may lead to heating, so coal can be left along floor, but there will be problem of maintaining horizon along coal floor.



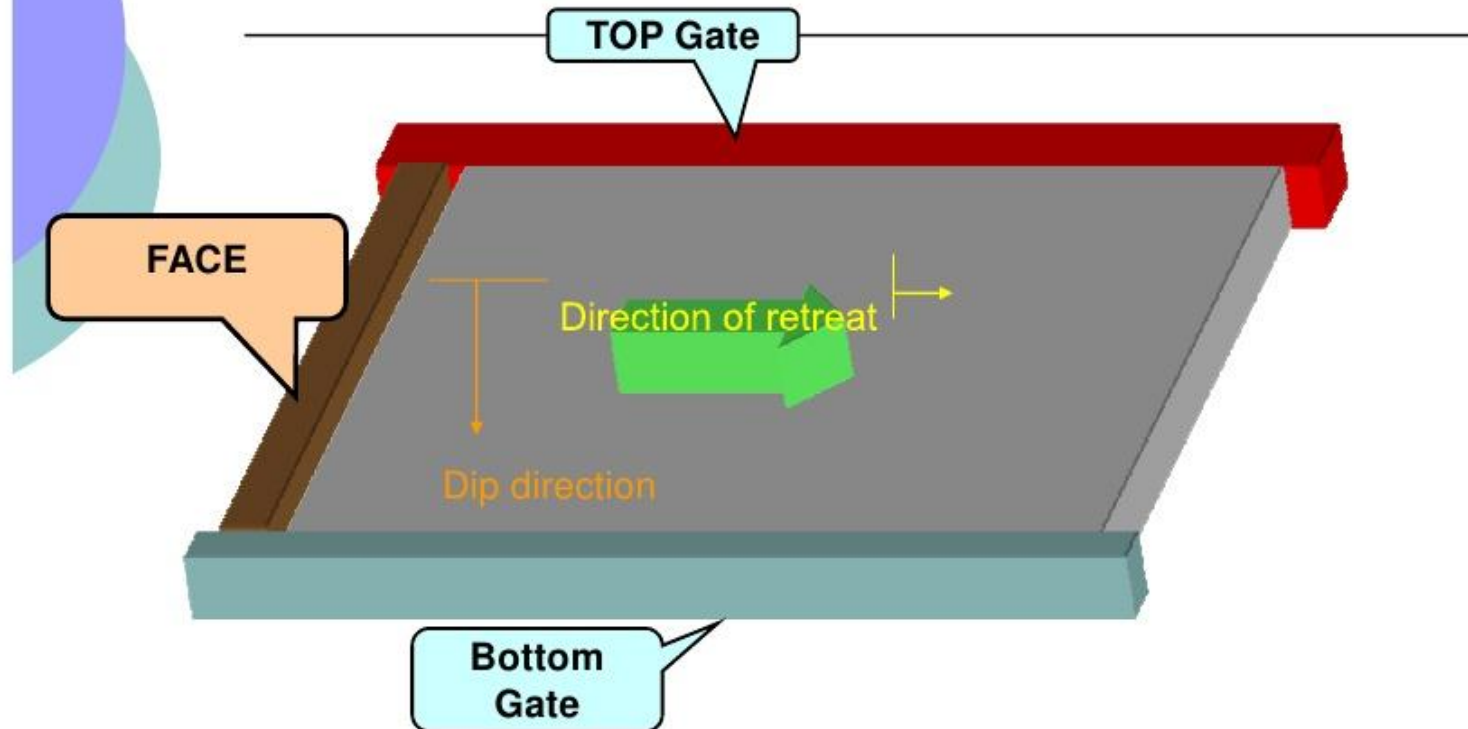
## Factors which constrain longwall block

- ▶ Size and shape of coal lease, location of existing workings (including those in under or overlying seams) and desirability of avoiding geological structures can all constrain the size of longwall blocks;
- ▶ Wider a face the slower the face advance rate for a given cutting rate, which can have bad effect on strata control;
- ▶ Generally the wider the face the more difficult it becomes to control alignment and to steer;
- ▶ Long blocks require large conveyor installations, sometimes making an additional conveyor drive (usually a tripper drive) necessary;
- ▶ Larger the blocks (length and width), more arduous the ventilation duty, particularly for lower seam heights.
- ▶ Long gate roads affect both longwall and development ventilation;
- ▶ Higher the production rate, greater the gas make which the ventilation and gas drainage system will have to accommodate;

# Factors which constrain longwall block

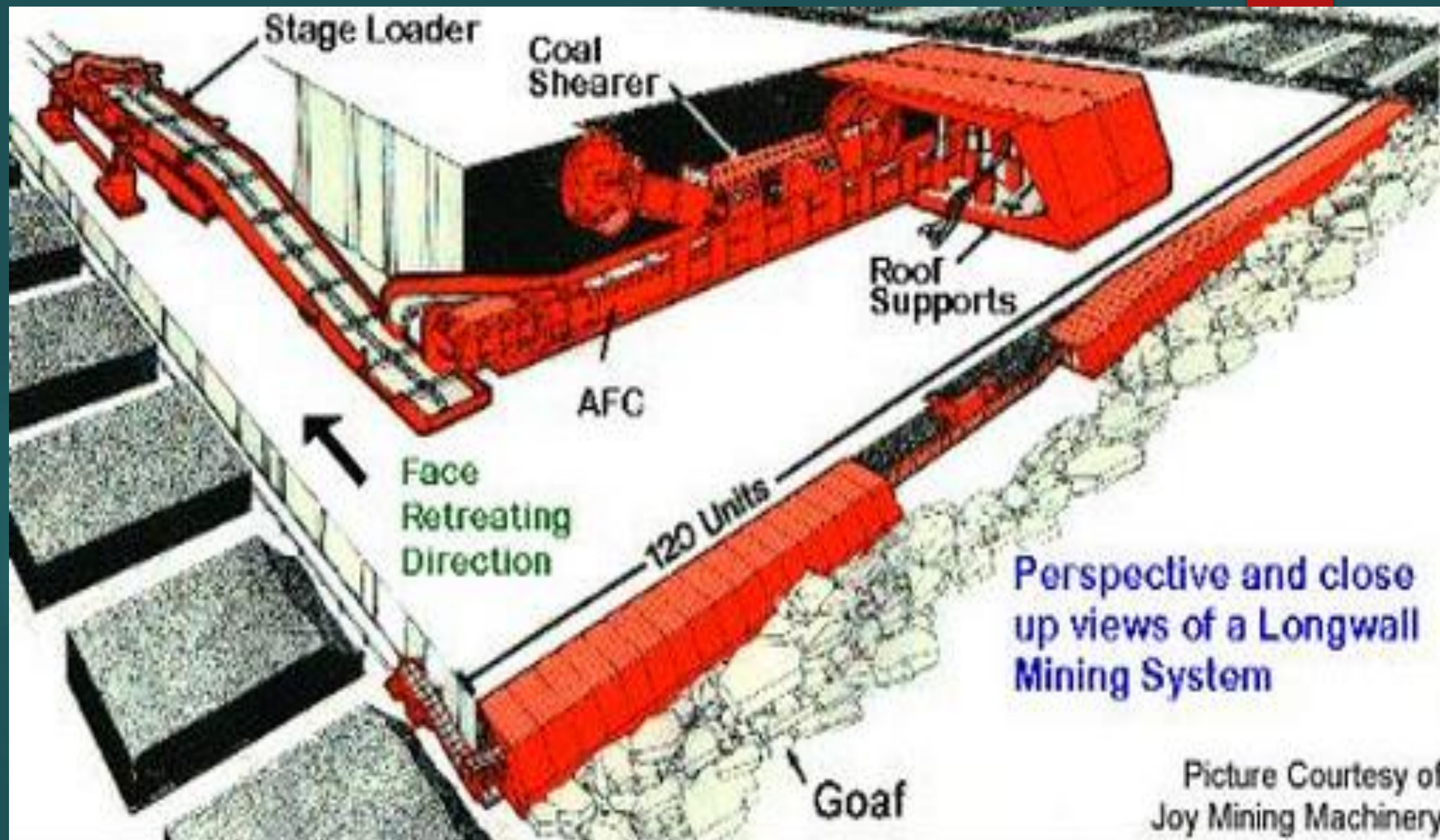
- ▶ Larger the block size, more arduous are supply water, compressed air
- ▶ Larger the block, possibility of major overhauling of major equipment during block life.
- ▶ Need of spare item to prevent the face to stand for long periods while carrying overhauls.
- ▶ Strata control may be critical problem during inevitable standing time involved, even where spare equipment is available;
- ▶ Wider the block, higher the initial capital cost involved.
- ▶ It is an option to increase block widths incrementally over several blocks to spread this cost, but certain items would have to be specified for the later higher duty when first purchased; and
- ▶ AFC chain size is also a consideration here and in some mines heat produced can be restrictive.

# LONGWALL GENERAL LAYOUT



- Longwall mining method includes drivage of two long roadways in coal and joining them at the end by a perpendicular drivage forming a face.





# Access to Longwalls

## ▶ Shaft

- ▶ Good ventilation
- ▶ Limited Hoist capacity
- ▶ Difficult machine transportation

## ▶ Inclines

- ▶ Transport of man, material, coal and machines – no restriction
- ▶ Greater maintenance

## ▶ Punch Entry from Opencast

- ▶ Easy entry / Follow coal seams, easy Transportation
- ▶ Multiple UG conveyor from different panels
- ▶ No. of entries can be made from the bottom of OC
- ▶ Stability of high wall and problem of inrush of water

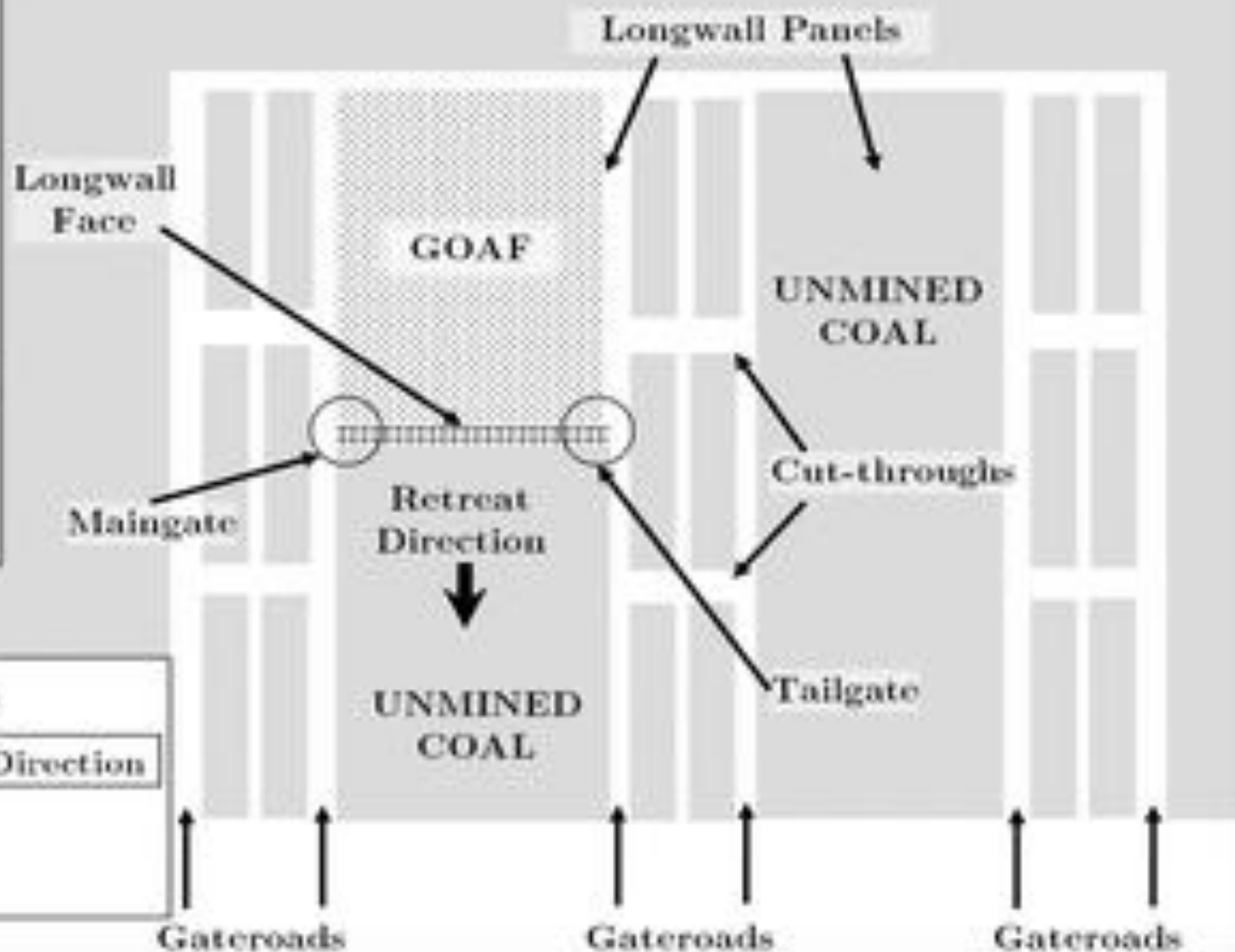
# Types of Longwalls

- ▶ Advancing
- ▶ Retreating
- ▶ Single / Double unit



### Longwall Process

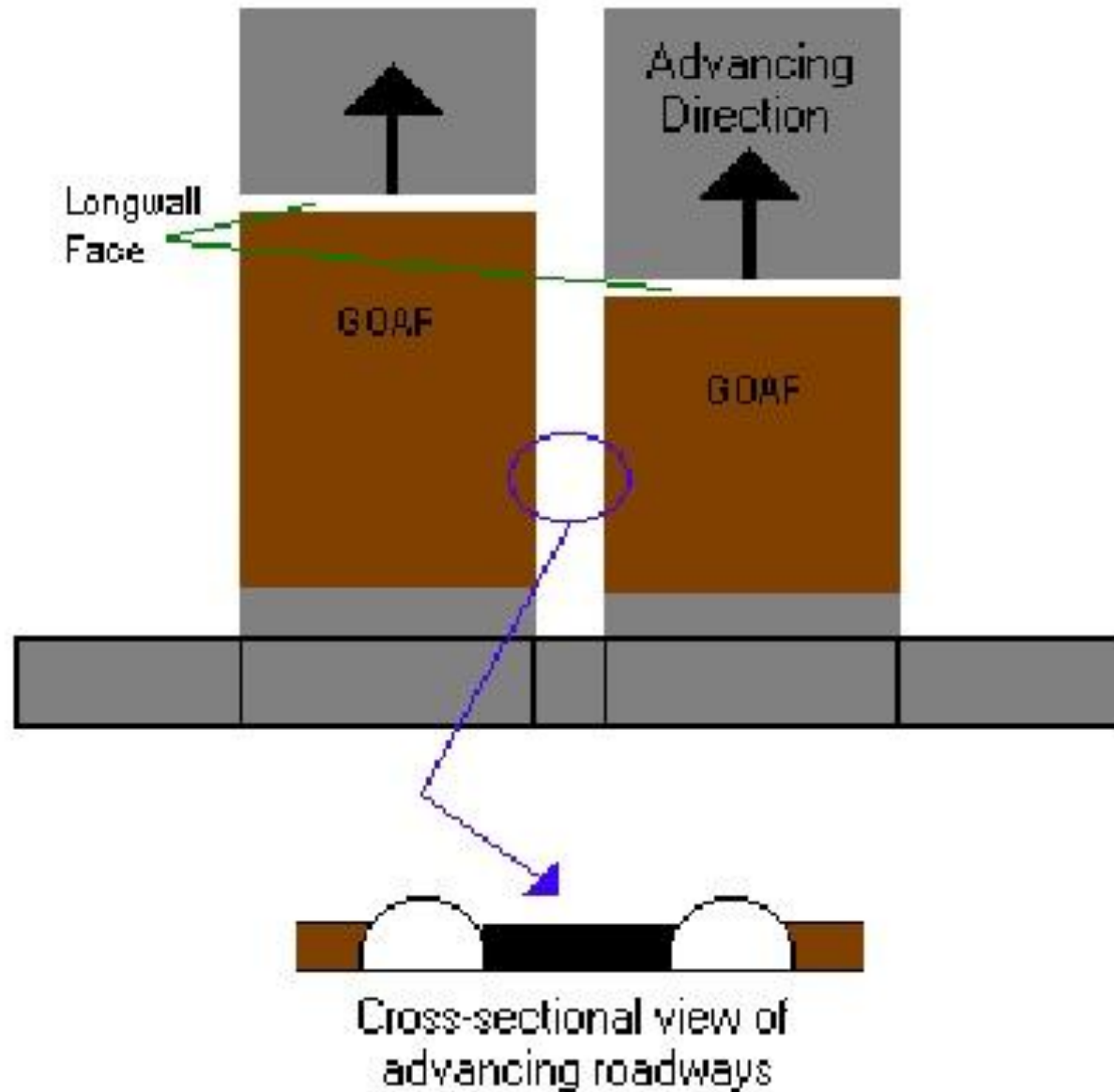
Coal is mined from the longwall face and conveyed out to the surface via the maingate gateroad. The longwall face mining equipment moves forward into the void left by the extracted coal, and the roof collapses behind the equipment as the face retreats (leaving the unstable goaf region behind).



# Advancing Longwall Faces

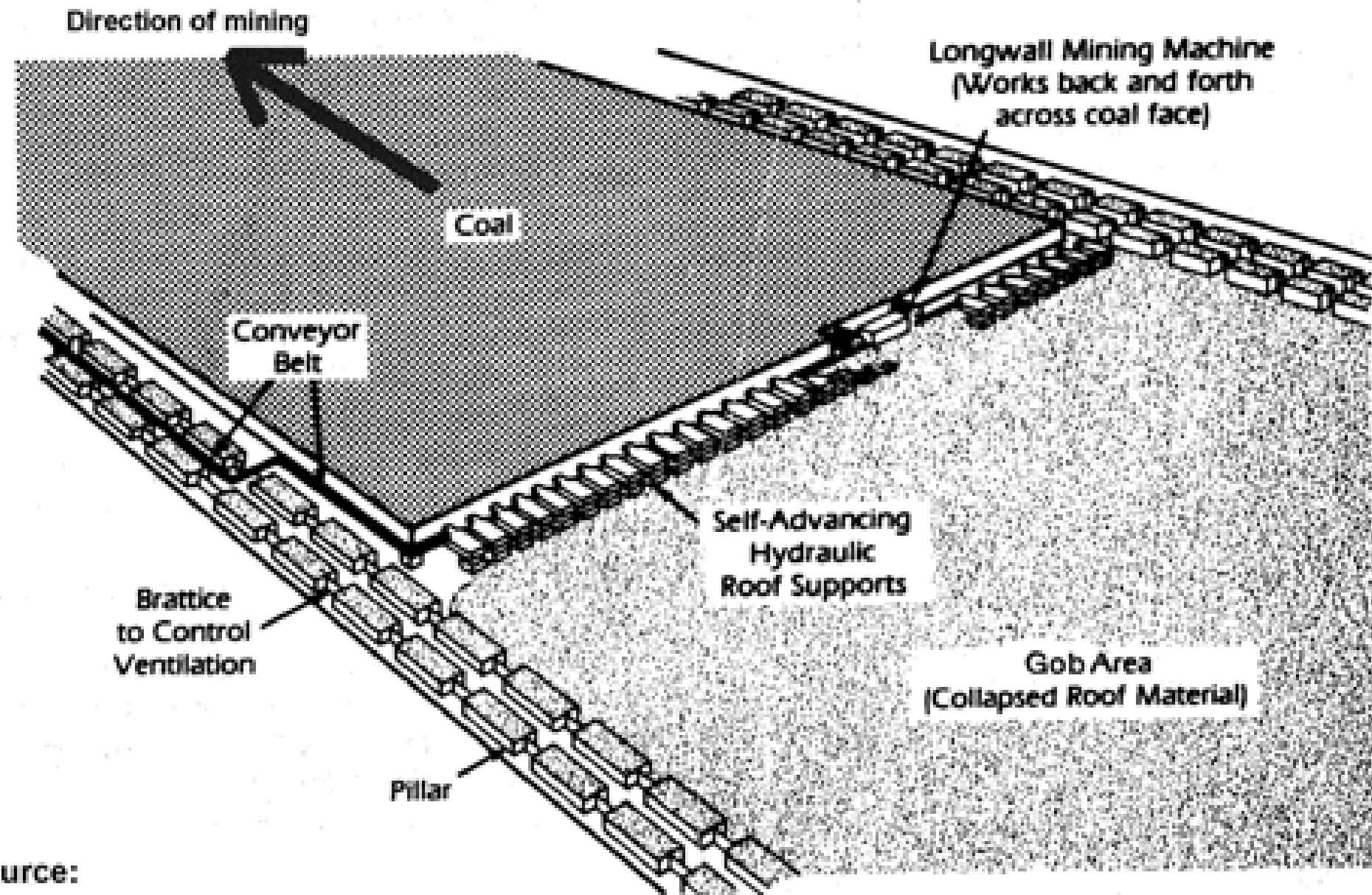
Face advances away from roadway

Main Roadways





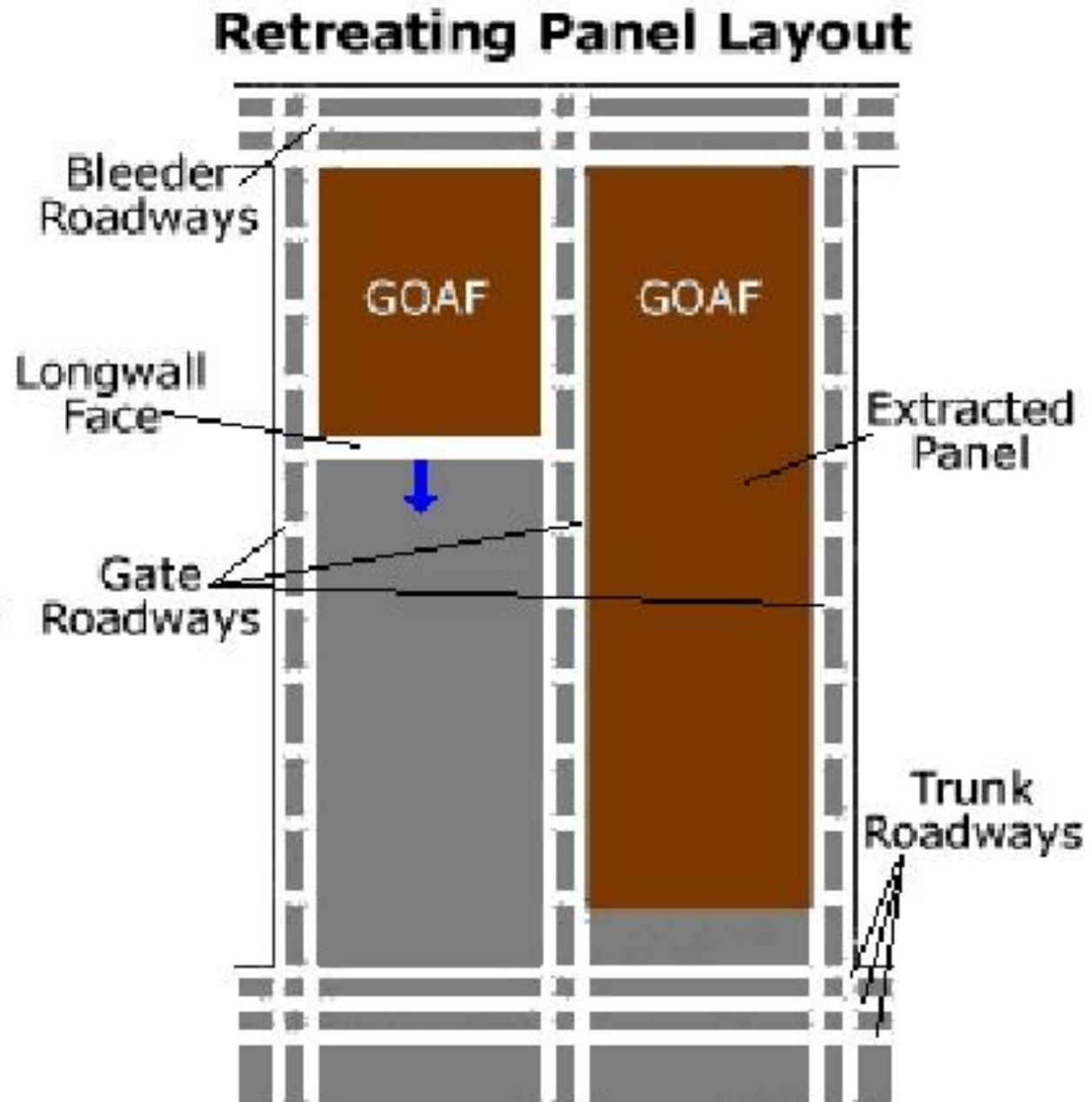
## Longwall



Source:  
Energy Information Administration

Two parallel tunnels are driven away from the main roadway and then connected at the longwall face to create a panel. The face works back – retreats - to the main roadway.

## Retreat Longwall Faces



# ADVANCING AND RETREATING LONGWALLS

## Comparison of Advancing and Retreating Longwalls

- ▶ Longwall block is surrounded by roadways before the longwall starts so knowledge of strata conditions is much better
- ▶ Area is geologically proved in advance during development of gateroads
- ▶ With retreat longwall mining, additional gateroads or bleeder roadways behind the goaf area can be developed for ventilation by the development unit if required. Such additional roadways are much more difficult to mine with an advancing longwall
- ▶ There are more options for ventilation/gas control using additional roadways at the limit of the block
- ▶ Gateroads are supported by solid coal atleast in one side
- ▶ Gateroad maintenance is minimum
- ▶ Production during development is substantial and it gives revenue

# ADVANCING AND RETREATING LONGWALLS

## Comparison of Advancing and Retreating Longwalls

- ▶ Gate road formation is remote from face operations
  - ▶ less congestion at face ends,
  - ▶ less supplies into longwall face area,
  - ▶ face not held up waiting for gate road preparation or vice versa,
  - ▶ no problems of dust production from gate road workings affecting longwall personnel).
- ▶ No gate or roadway side packs required, so less supplies overall
- ▶ Gas drainage of adjacent blocks can be carried out starting during development;
- ▶ with longwall advancing the drilling can only be done behind the face after longwall extraction, allowing less drainage time before the next block commences production

# ADVANCING AND RETREATING LONGWALLS

## Comparison of Advancing and Retreating Longwalls

- ▶ Advancing longwall gateroads typically require extensive maintenance to maintain the roadway cross-section (roof and floor brushing) during the life of the longwall block, whereas retreat longwall gateroads are allowed to collapse behind the retreating face
- ▶ Face Ventilation may be sluggish due to lag in construction of packwall behind the advancing face
- ▶ Ventilation in advancing headings is poor as it draws part of the air after ventilating the longwall face
- ▶ Chance of spontaneous heating is more

# Advantage of advancing

- ▶ Production can begin earlier as the mine does not have to wait for the gate roads to be developed before longwall production can commence (provided development rates are adequate this should only apply for the first longwall in a mine)
- ▶ It provides an opportunity for disposal of stone which has to be excavated into gate side packs (this benefit is probably more than offset by the costs involved in pack construction)
- ▶ Same system of coal evacuation, ventilation can be used for advancing gate-roads
- ▶ Supervision is concentrated



# Longwall development

## ▶ Gate road development

### ▶ No of entries

- ▶ Single entry

- ▶ Multiple entry

- ▶ Length of drivage

- ▶ Dimension of galleries

- ▶ Alignment / Orientation of gate roads

  - ▶ Face along dip-rise, advancing along strike

  - ▶ Face along strike and advance from dip to rise

  - ▶ Face along apparent dip

# Longwall development

## ▶ Gate road development

### ▶ Difficulties in gate road development

#### ▶ Stability

- ▶ Size of chain pillar

- ▶ Roof condition

#### ▶ Ventilation

#### ▶ Dust

#### ▶ Gas

#### ▶ Temperature

#### ▶ Humidity

#### ▶ Transportation of man, material and coal

# Equipment for gate road development

- ▶ Road header
  - ▶ Bridge-belt conveyor
  - ▶ Extensible belt conveyor
  - ▶ Trunk conveyor
- ▶ Continuous miner
  - ▶ Shuttle car
  - ▶ Bolter
  - ▶ Feeder Breaker
  - ▶ Extensible belt conveyor



# Roadway development by RH Operations

## Steps

- ▶ Coal getting
  - ▶ Sumping
  - ▶ Cutting – cutter drum
  - ▶ Water spraying – pick face flushing
  - ▶ Loading – Gathering arm loader
  - ▶ Transferring onto bridge conveyor
  - ▶ Bridge conveyor to extensible conveyor
- ▶ Supporting – roof bolting
- ▶ Ventilation duct extension
- ▶ Pumping

# Roadway development by CM

## Steps

- ▶ Coal getting
  - ▶ Sumping
  - ▶ Cutting – cutter drum
  - ▶ Water spraying – pick face flushing
  - ▶ Loading – Gathering arm loader
  - ▶ Transferring onto central conveyor
  - ▶ Central conveyor to shuttle car

# Roadway development by CM

## Steps

- ▶ Supporting – roof bolting by separate bolter or bolter-miner
- ▶ Making cut through between two gate-roads
- ▶ Ventilation duct extension
- ▶ Shifting of auxiliary fan beyond last cut through
- ▶ Location of fan – forcing in intake side/ exhaust in return side
- ▶ Quantity shall not be less than double the fan capacity
- ▶ Pumping
- ▶ Cleaning of face by LHD for place changing method with CM



# Roadway development by CM

## Steps for place changing method

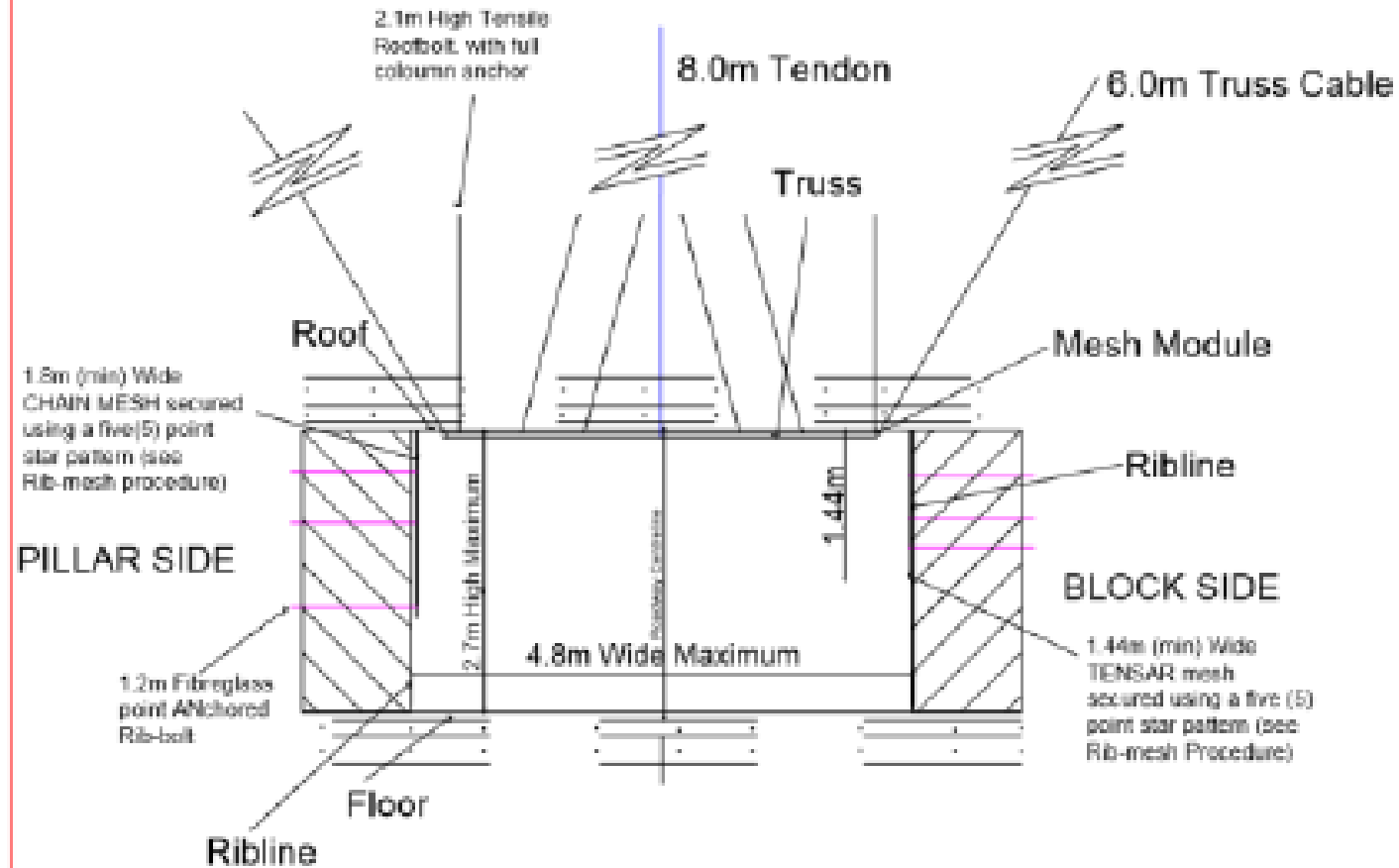
- ▶ Cleaning of face by LHD for place changing method with CM
- ▶ Face advance in Face 1
- ▶ Retreat from face 1 to other face 2
- ▶ Bolting in Face 1 by separate bolter
- ▶ Cutting and face advance in Face 2
- ▶ Feeder breaker and conveyor in one gate road
- ▶ After each cycle – swapping of CM and Bolter

# Roadway development by CM

## Steps for in-place method

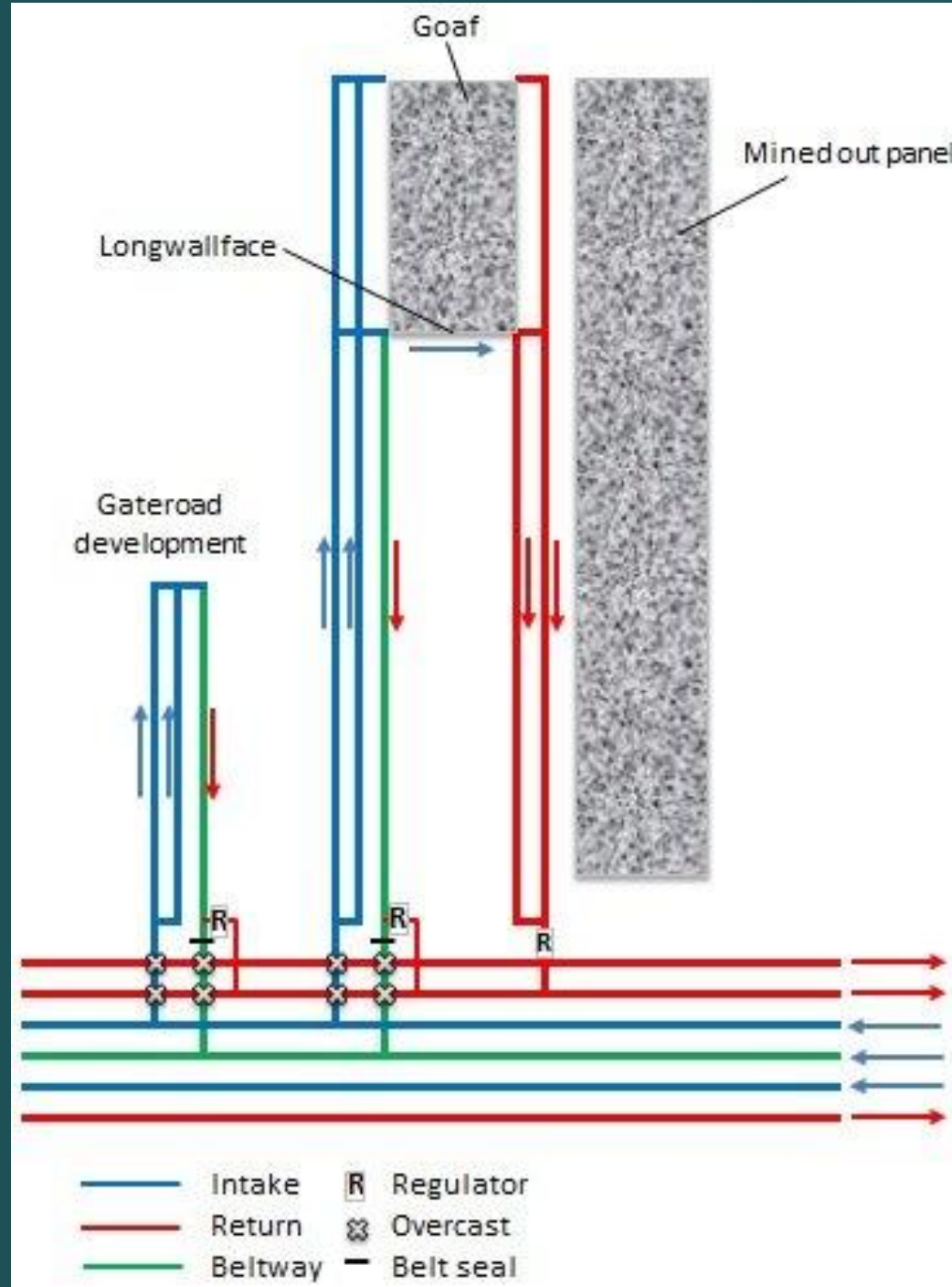
- ▶ Face advance including support installation in Face 1
- ▶ Retreat from face 1 to other face 2 through last cut through
- ▶ Cutting and face advance including support installation in Face 2
- ▶ Feeder breaker and conveyor in one gate road

# Support in gateroads



# Ventilation

- ▶ Auxiliary Fan
  - ▶ Forcing
  - ▶ Exhaust





# Longwall extraction

- ▶ Face equipments
  - ▶ Shearer
  - ▶ Powered supports
  - ▶ AFC
  - ▶ Stage loader
  - ▶ Control unit
  - ▶ Energy train
  - ▶ Monitoring arrangements



# POWERED SUPPORTS

- ▶ Hydraulic powered roof supports are self advancing structures which are interconnected along the length of the longwall face.
- ▶ Each unit along the line is also connected to each pan of the AFC.

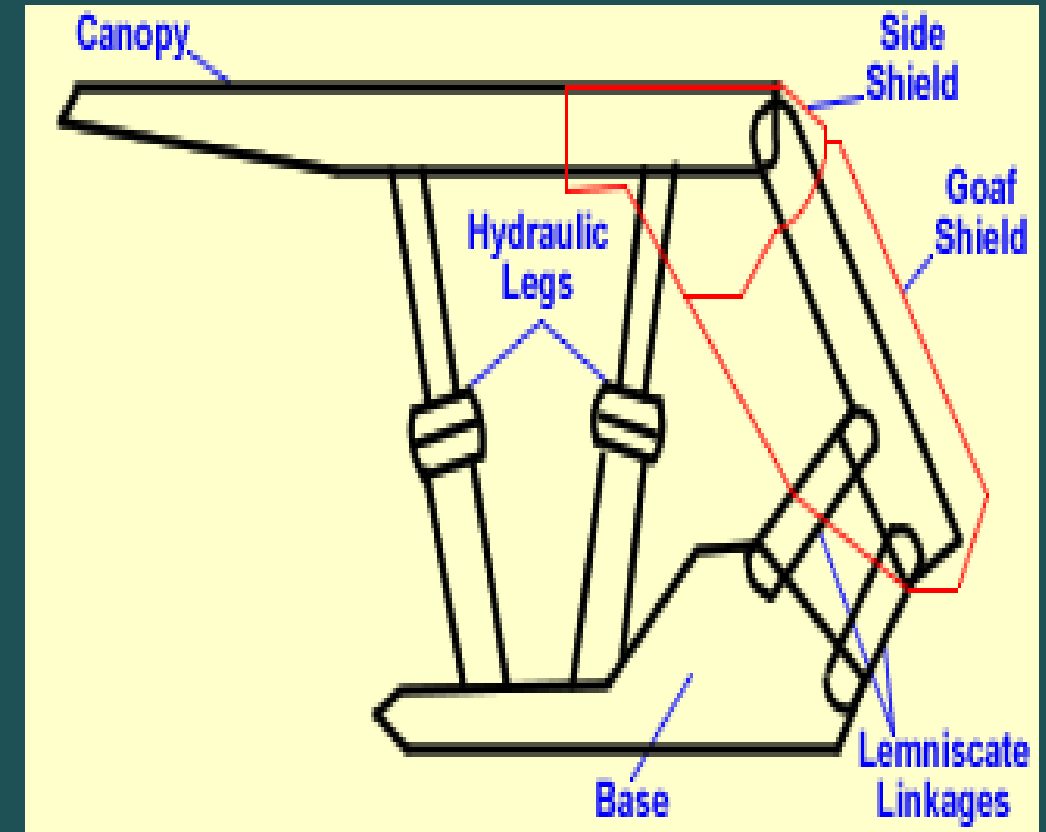
# POWERED SUPPORTS

The function of the powered supports are to:

- ▶ To control strata deformation, fracture and movement around the coal face,
- ▶ To maintain a safe and coal producing working environment,
- ▶ To limit the amount of roof to floor convergence,
- ▶ To prevent broken rock from entering the work area, and
- ▶ To secure and advance all plant on the coal face including the roof supports.

# POWERED SUPPORTS

- ▶ Each support structure consist of a roof canopy connected to a base via a shield and lemniscate linkages.
- ▶ Side shields mounted to the canopy prevents excessive debris falling into the work space during support advancement.
- ▶ The hydraulic legs of a each type of support are inclined or splayed at an angle.
- ▶ This is to prevent the legs from buckling under the immense loads that they support.



# Types of powered supports

- ▶ Frame type supports
- ▶ Chock supports
- ▶ Shield supports
- ▶ Chock shield supports

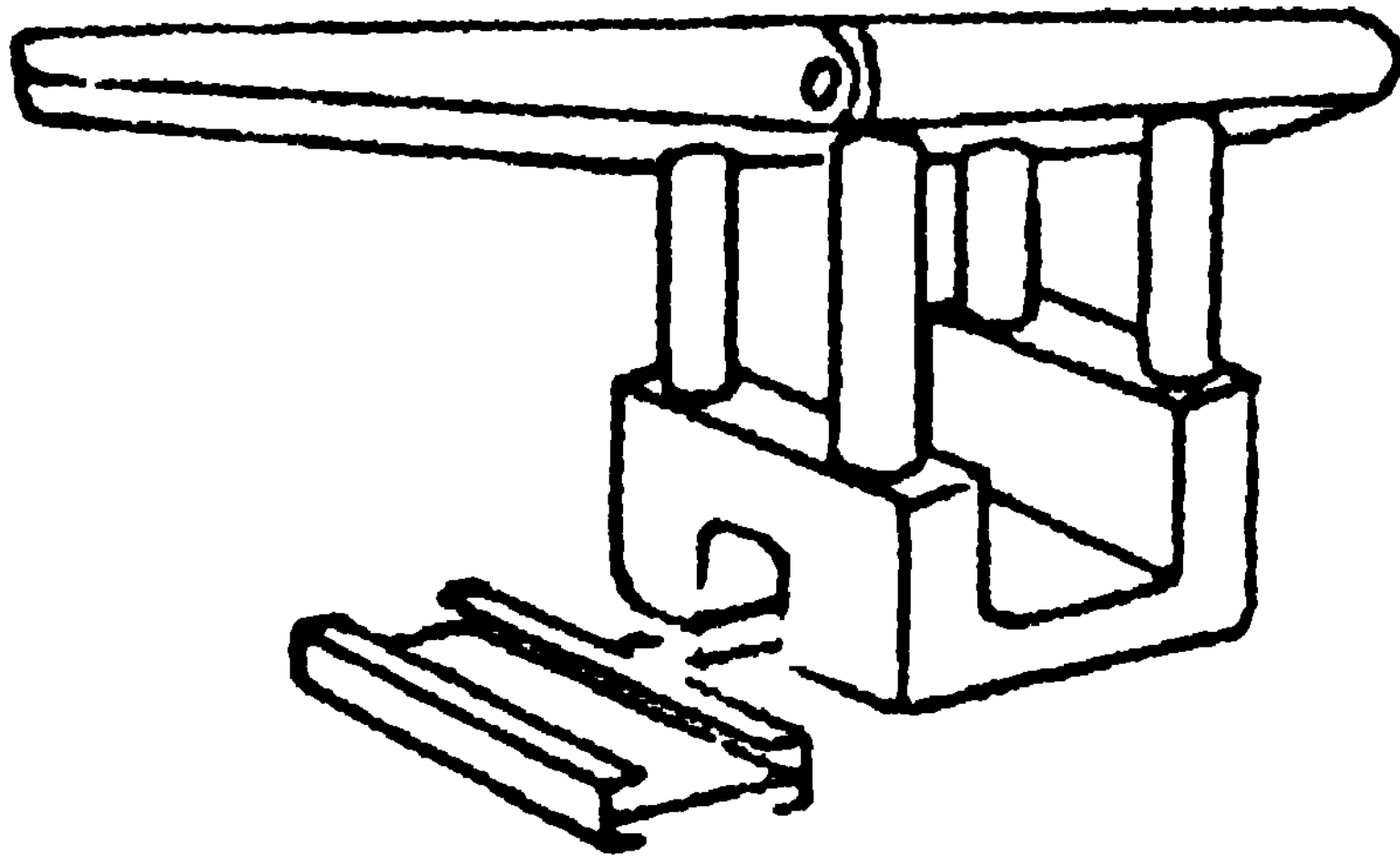
# Frame supports

- ▶ First generation supports
- ▶ Walking type – supports are advanced by walking on legs
- ▶ Legs have individual base plates
- ▶ Two legs have one common canopy
- ▶ Two adjacent units are connected with rams
- ▶ Flat back shield – not lemniscate
- ▶ Not very stable
- ▶ Simple to operate

# Chock supports

- ▶ Chock supports have 3 to 6 hydraulic legs
- ▶ 4 legged chocks are most common
- ▶ All 4 legs are connected to the canopy of the support.
- ▶ All legs mounted on a common base
- ▶ Legs of a chock shield support are either vertical or inclined but the majority of chock shields have inclined legs as this provides better support pressure to the canopy.
- ▶ Whole unit moves at a time
- ▶ Provide more support resistance
- ▶ Not very stable from back thrust





*Four-leg chock*

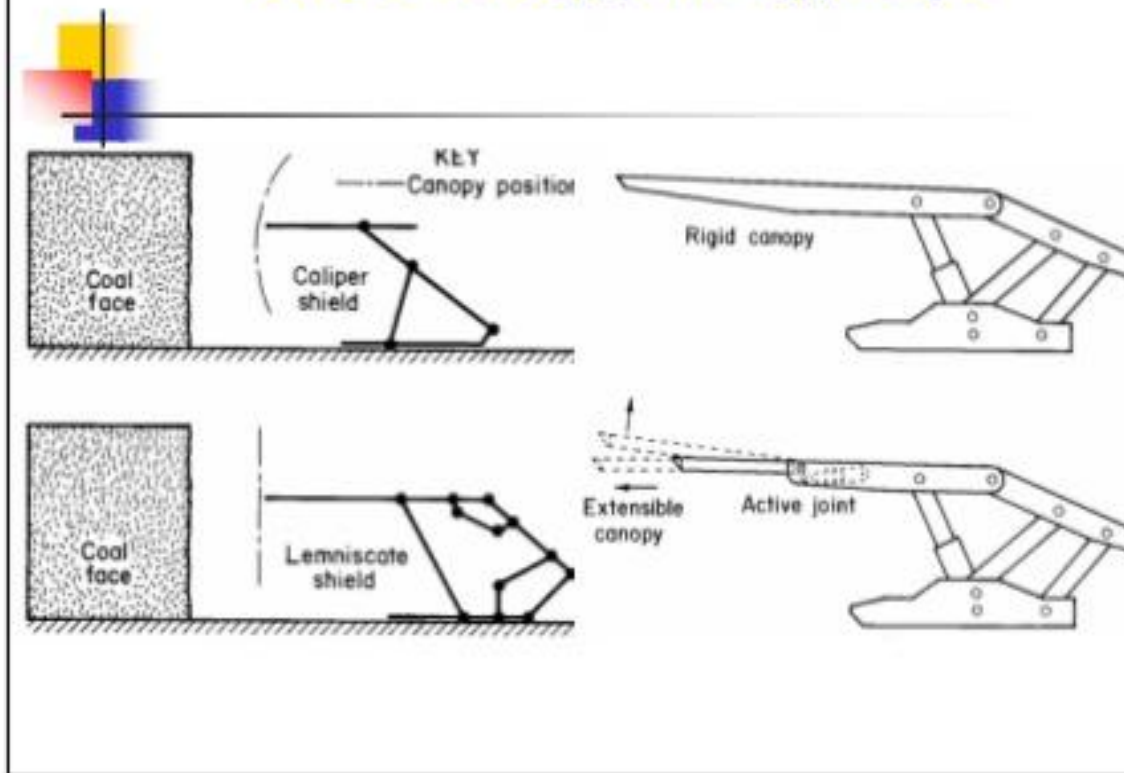
# Shield supports

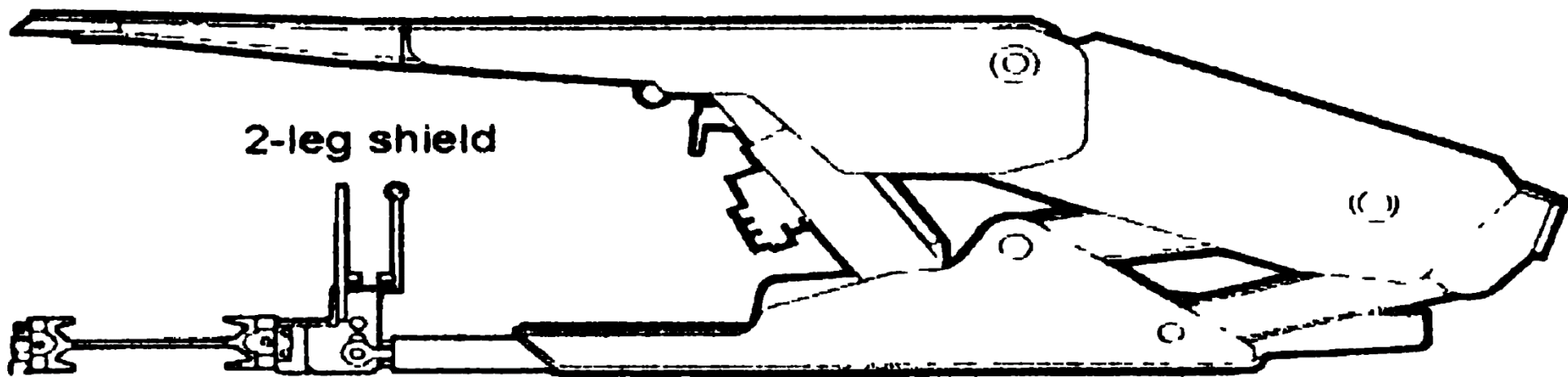
- ▶ May be two legged or four legged
- ▶ Have shield at the back with lemniscate linkages
- ▶ Caving shield, which are in general inclined, are hinge jointed to the canopy and, together with lemniscate assembly, to the base making the shield kinematically stable support, a major advantage over frame and chock supports.
- ▶ The caving shield also seals off the gob and prevents rock debris from getting into the face side.

# Shield supports

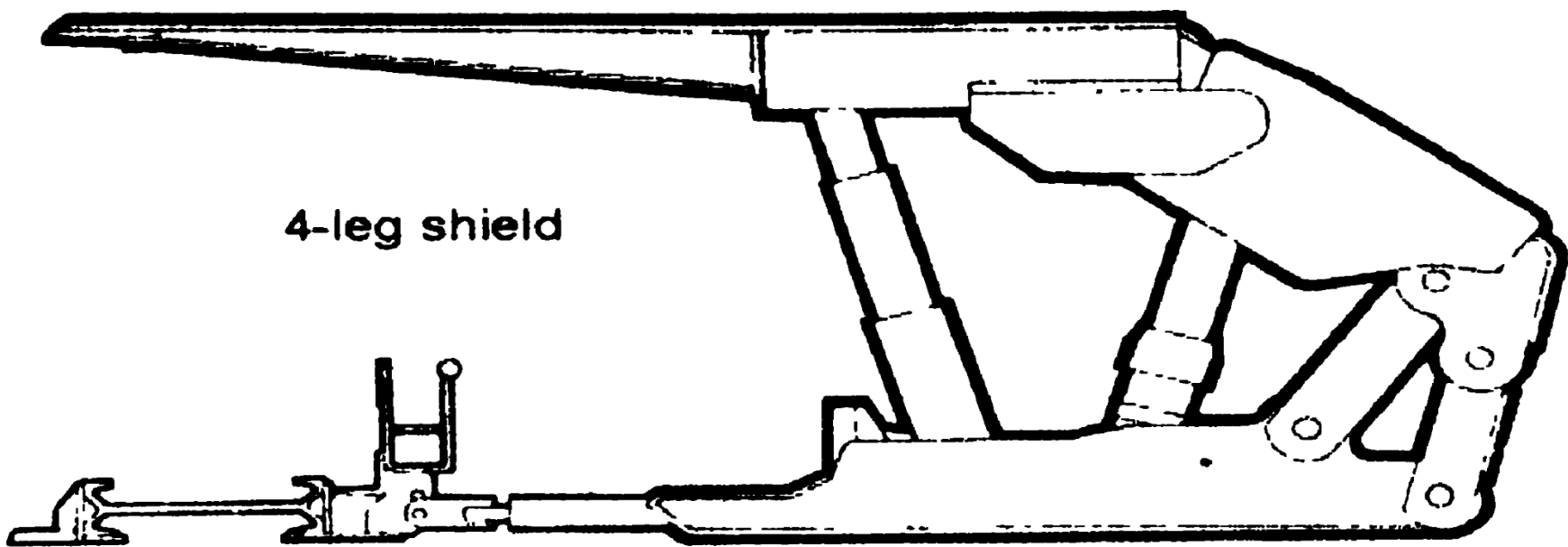
- ▶ Hydraulic legs in shields are generally inclined to provide a larger working height range and more open space for traffic.
- ▶ Because the canopy, caving shield and base are interconnected, shield can well resist horizontal force without bending legs.
- ▶ Roof support capacity of the shield varies with mining height as only vertical component of the leg pressure is available for supporting roof.
- ▶ At least two legs connected between base frame and shield
- ▶ In four legged shield two sets of legs inclined forward into the canopy and the other two inclined backwards and connected to the goaf shield

## Powered Roof Support Canopy Designs





2-leg shield



4-leg shield

# 4-legged Shield supports

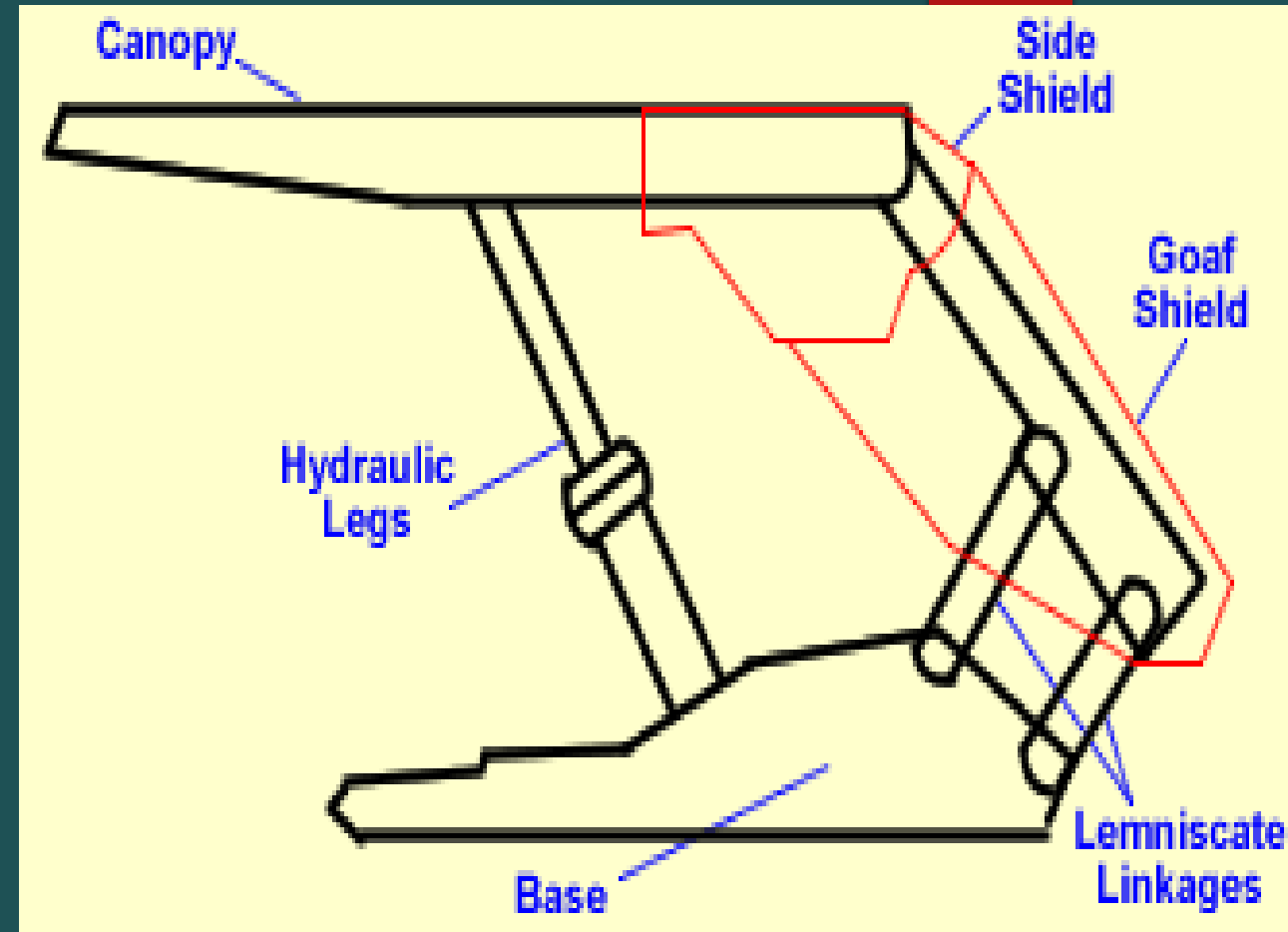
The main features of the 4 leg shield are:

- ▶ The front legs connect the base and the canopy and the rear legs connect the base and the top shield member
- ▶ The top canopy is hinged to the main rear canopy to allow the canopy to articulate along the line of the face
- ▶ The base is connected to the rear shield with a lemniscate linkage

# Shield supports

The main features of the 2 leg shield are:

- ▶ two legs are connected into the canopy at an inclined angle and the supports are usually operated in Immediate Forward Support (IFS) mode.
- ▶ support density is not uniform throughout the working range
- ▶ lemniscate linkage ensures that the front of the top canopy maintains a constant distance from the coal, between the fully closed state and fully open position
- ▶ Full use of the reverse mounted ram
- ▶ Side ram shields ensures good flushing protection





# Why two legged support

## 1. Simplicity

- 2-legged shields have only half as many hydraulic components (Legs, cables, hoses, valves)
- Walkway cleaner and spacious
- Much simpler to operate
- Require less maintenance

## 2. High capacity

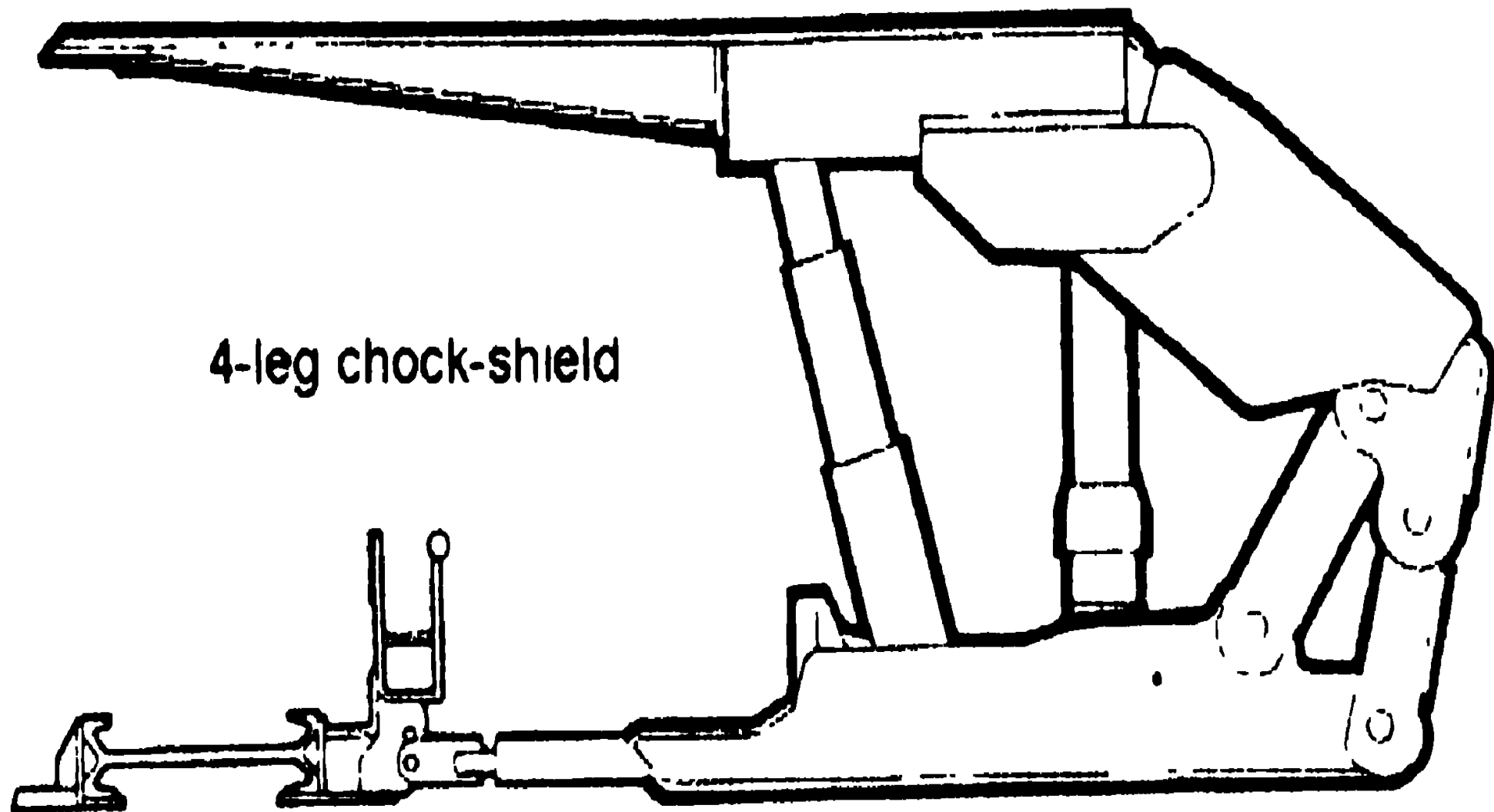
- With technological advancement for manufacturing larger leg cylinders, 2-legged legs also offer high support capacity

## 3. Improved design features

- Positive base lifting device

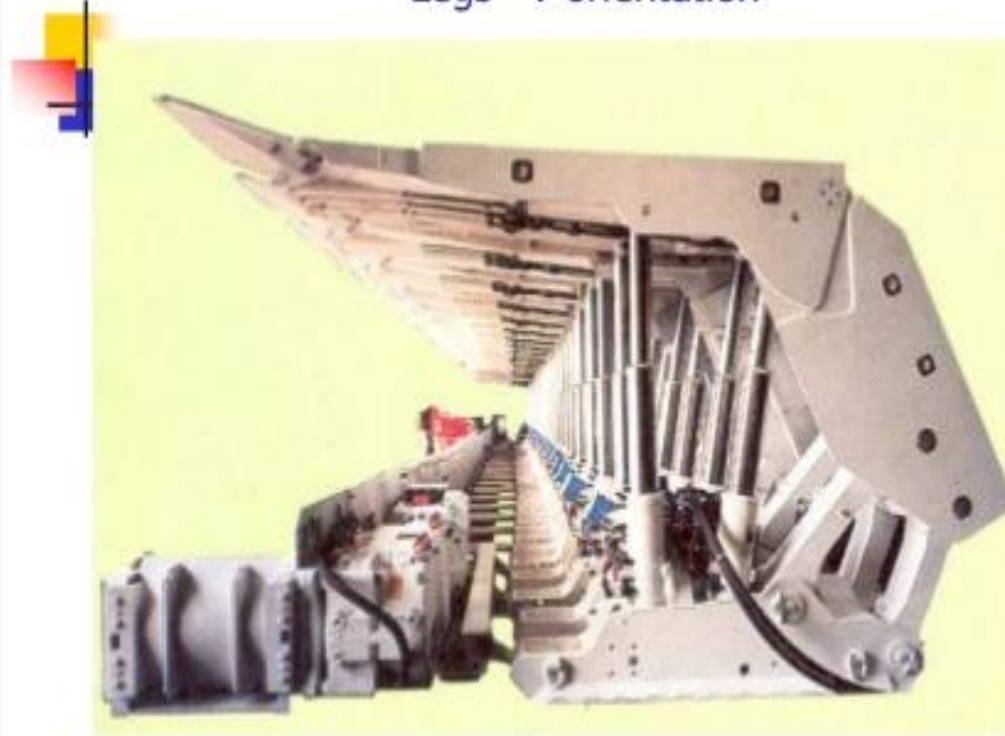
# Chock shields

- ▶ Combination of chock and shield supports
- ▶ It is a shield with four legs between base and canopy
- ▶ The legs of a chock shield support are either vertical or inclined but the majority of chock shields have inclined legs as this provides better support pressure to the canopy.
- ▶ Four legs act directly onto the canopy
- ▶ Yield load is constant throughout the height range
- ▶ Top canopy can be either rigid or articulated
- ▶ Base unit is rigid but has no leg mountings
- ▶ Suitable for hard roof



4-leg chock-shield

4 legged Chock Lemniscate Shield Support ,  
Legs -V orientation



### Caliper Shield Support



4x410 Tonne ,I.F.S , Chock Shield with rigid roof bar



I.F.S. support with rigid roof bar.

4x410 Tonne ,I.F.S , Chock Shield with articulated forward bar





# Shield operation mode

- ▶ Manual mode or Remote control (batch mode)
- ▶ In manual mode
  - ▶ each shield operated separately by operator from the adjacent support uni-directionally or bi-directionally
- ▶ In batch mode
  - ▶ Number of supports in a batch are advanced using a software program
- ▶ Shearer initiated control
  - ▶ In this mode, after passage of the shearer, the shield at a preset offset distance initiates the advance movement in sequence followed by bank push at a preset distance behind.

# Setting load and yield load

- While setting the support, fluid is fed into the bottom stage of the cylinder
- As fluid continues to flow, leg continues to rise until the canopy touches the roof
- After this the fluid in bottom stage chamber rapidly increases to the working pressure (pump pressure minus pressure loss) of the power pack
- Control valve is turned off and fluid is locked in
- This is setting pressure of the support
- After setting, roof usually continues to converge, compressing the fluid in leg cylinder, resulting into rising of fluid pressure in top and bottom stages of the cylinder
- As the roof convergence increases, the fluid pressure inside cylinder chamber continues to rise

# Setting load and yield load

- To protect the hydraulic cylinder and piston from damage, a maximum allowable pressure in the bottom stage of the cylinder, called yield pressure, is preset for each support, provided by a yield valve installed externally in the cylinder.
- When fluid pressure inside bottom stage chamber exceeds the yield pressure, the yield valve opens immediately and the fluid leaks out quickly, causing quick pressure drop below yield pressure.
- This process repeats again and again if roof continues to converge.
- Yield pressure is generally set at one fourth of the yield strength of the steel used for the cylinders and pistons

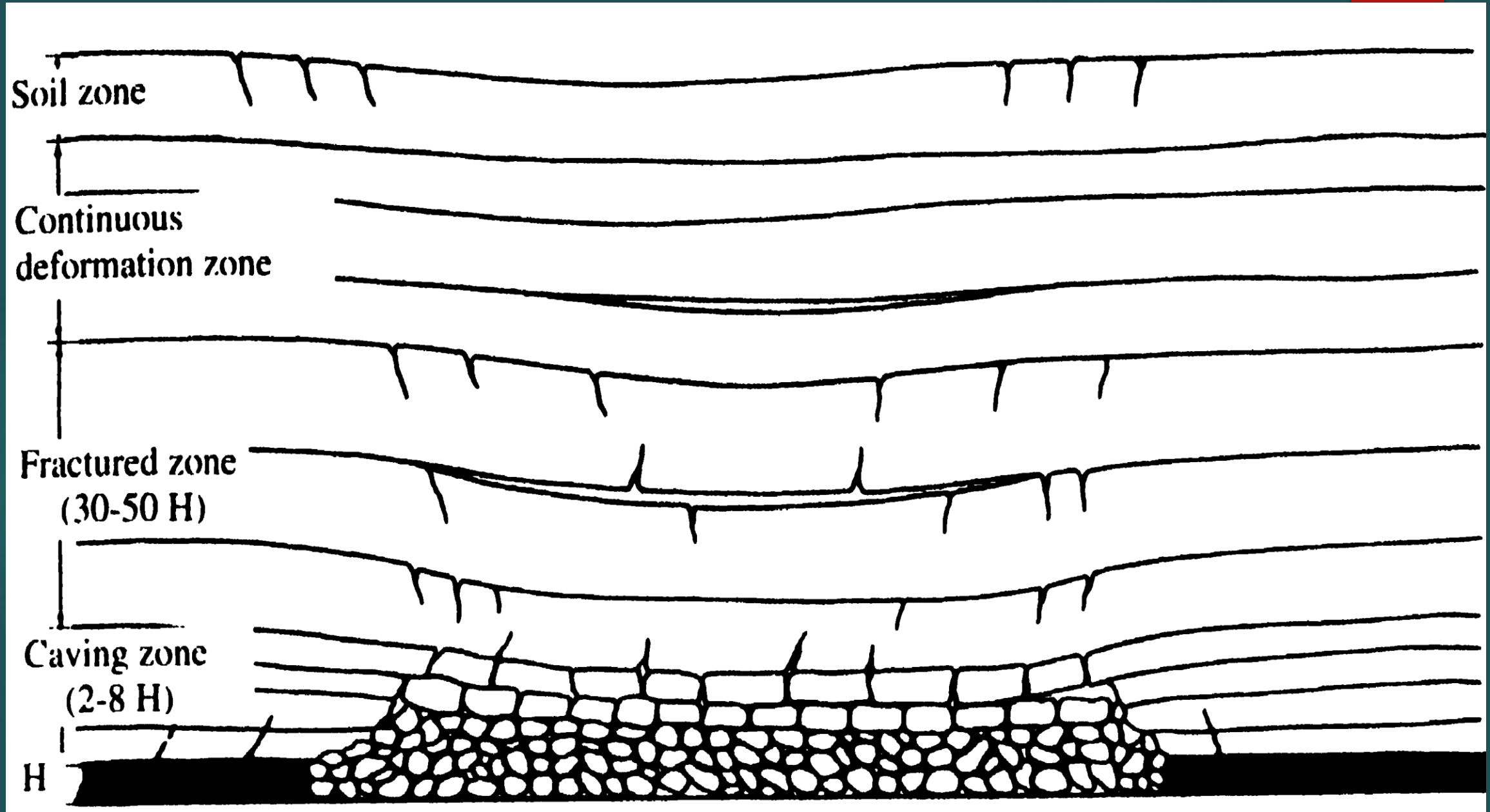
# Positive set and guaranteed set

- ▶ In an electro-hydraulic control system, two levels of setting a shield:
  - ▶ Positive set and Guaranteed set
  - ▶ Positive set is to ensure that hydraulic leg pressure reaches the rated setting pressure when the shield canopy is first raised to contact the roof
  - ▶ Guaranteed set is that after a positive set is implemented at the beginning of the cycle, a pilot signal will scan all the shields at a selected time interval. If the pressure in any shield is found to be below setting pressure, it will be raised to the rated setting pressure.

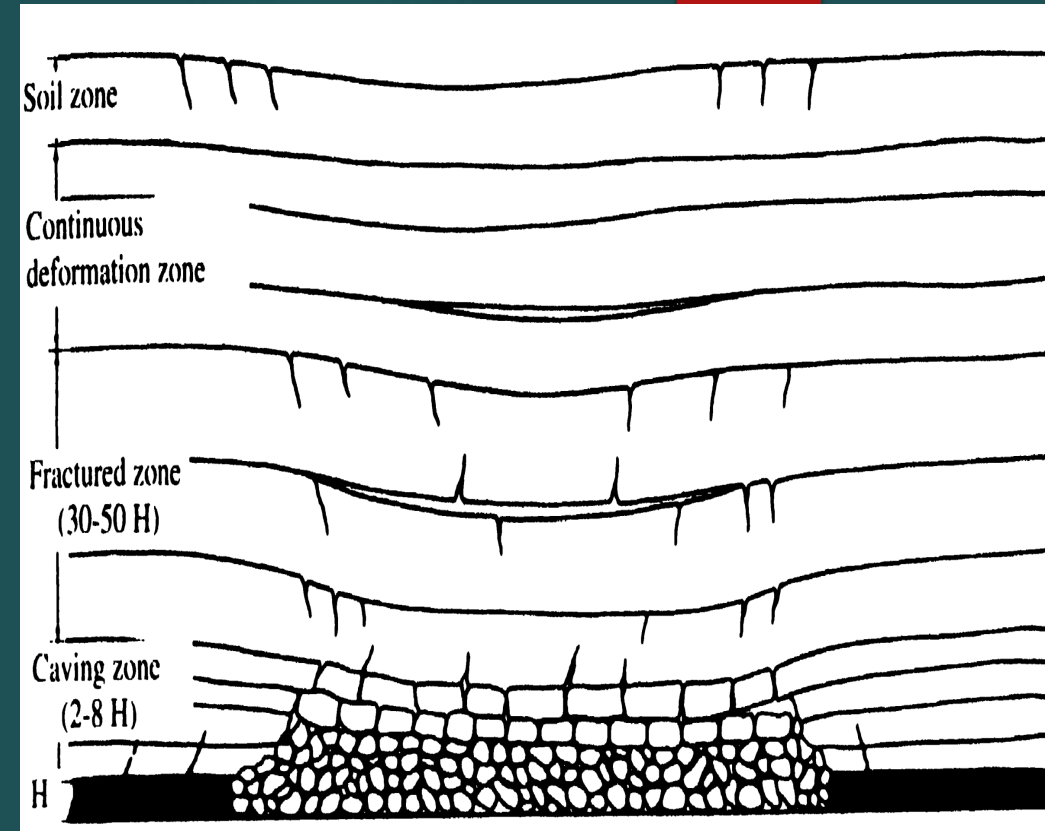
## Overburden movement resulting from longwall mining

- As the longwall panel is extracted slice by slice, surrounding strata are forced to move toward and attempt to fill the voids left by extracted coal.
- This process induces
  - movement of rock strata between roofline and surface, resulting in surface subsidence
  - abutment pressures on both sides of the panel
  - roof to floor convergence in gate roads and face areas

# Overburden movement resulting from longwall mining

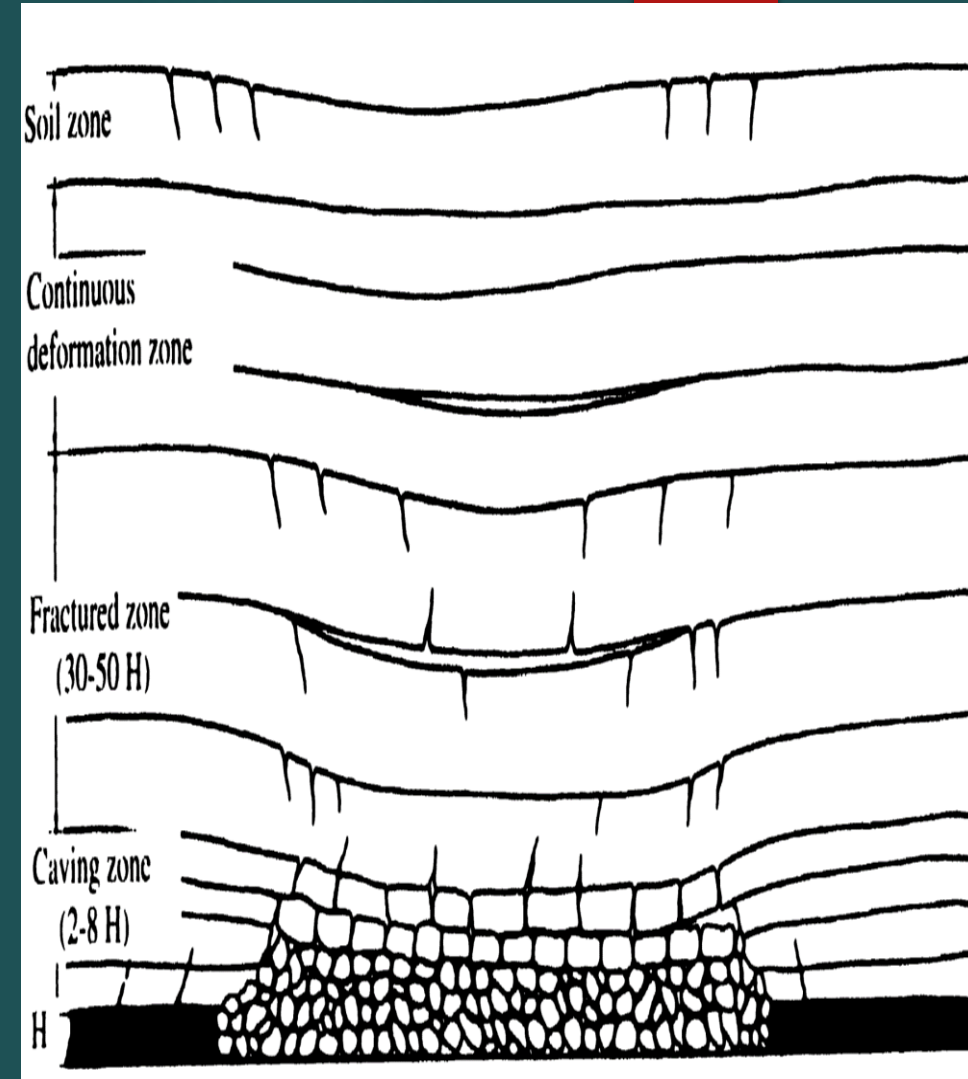


- ▶ When a longwall panel of sufficient width and length is excavated, the overburden strata is disturbed in order of severity from immediate roof towards surface
- ▶ The **caved zone**, which is immediate roof before it caves, ranges in thickness from two to eight times the height of extraction
- ▶ In this zone, the strata fall on the mine floor and in the process, broken into irregular but platy shapes of various sizes
- ▶ They are crowded in random manner
- ▶ Rock volume gets increased. The ratio called bulking factor
- ▶ It determines the caving zone

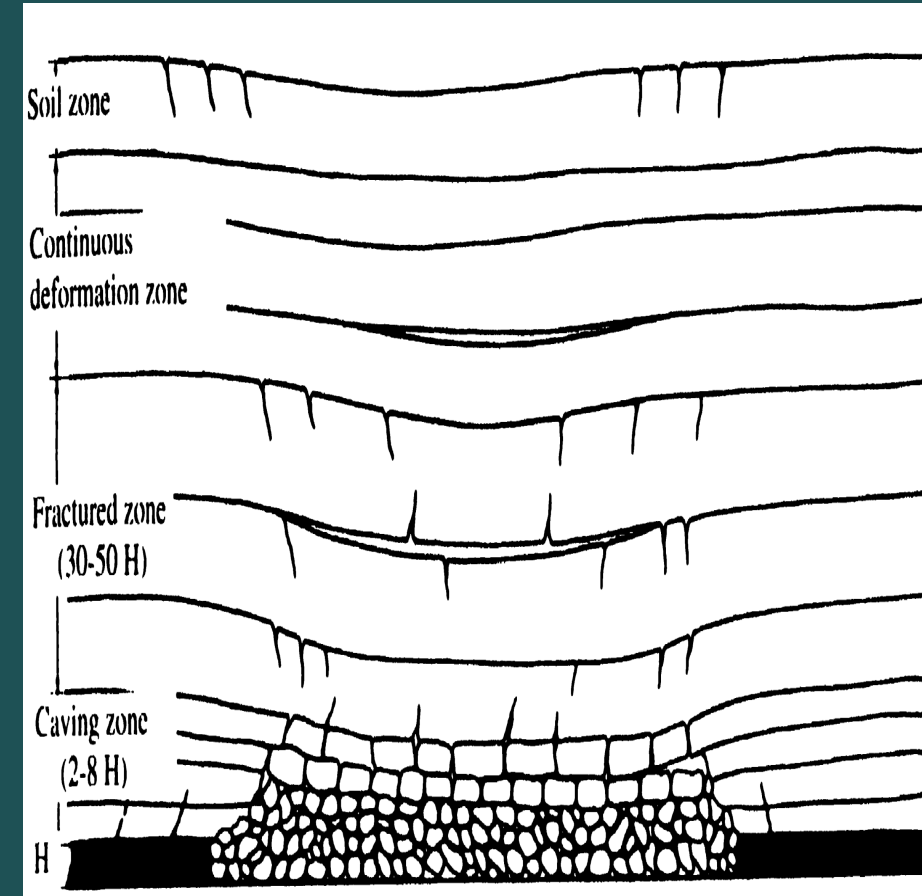




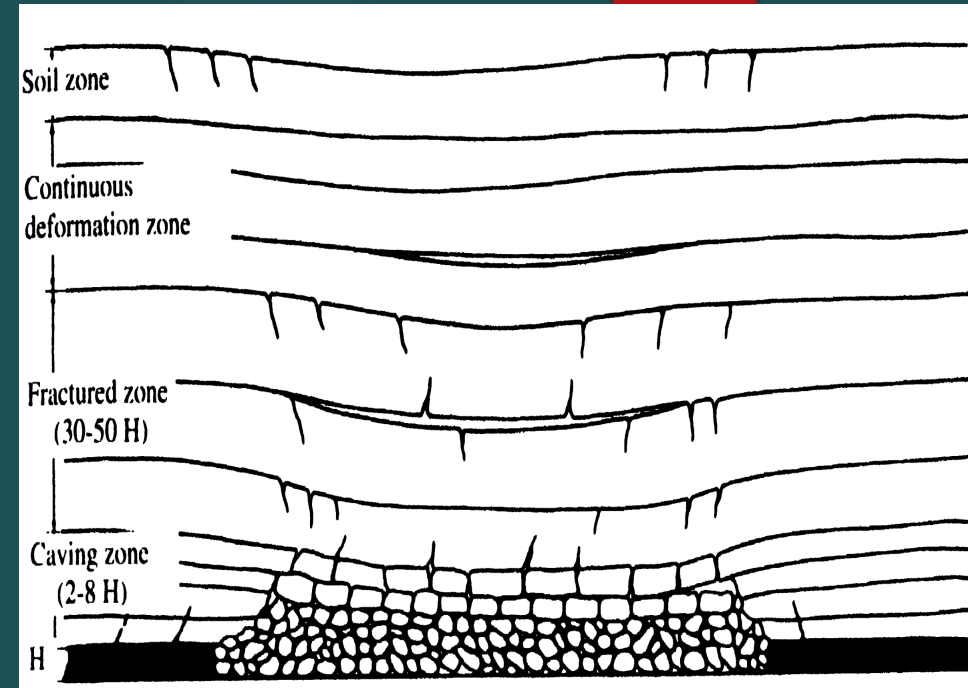
- Above the caved zone is **fractured zone**
- Strata broken into blocks by vertical or sub-vertical fractures and horizontal cracks due to bed separation
- Adjacent blocks in each broken stratum still maintain contact
- Thus there is a horizontal force transmitted through and remains in the strata
- These broken strata are called force transmitting beams
- Thickness of fractured zone is 28-52 times the height of extraction
- Combined thickness of the caved and fractured zones varies from 30 – 60 times the mining height.



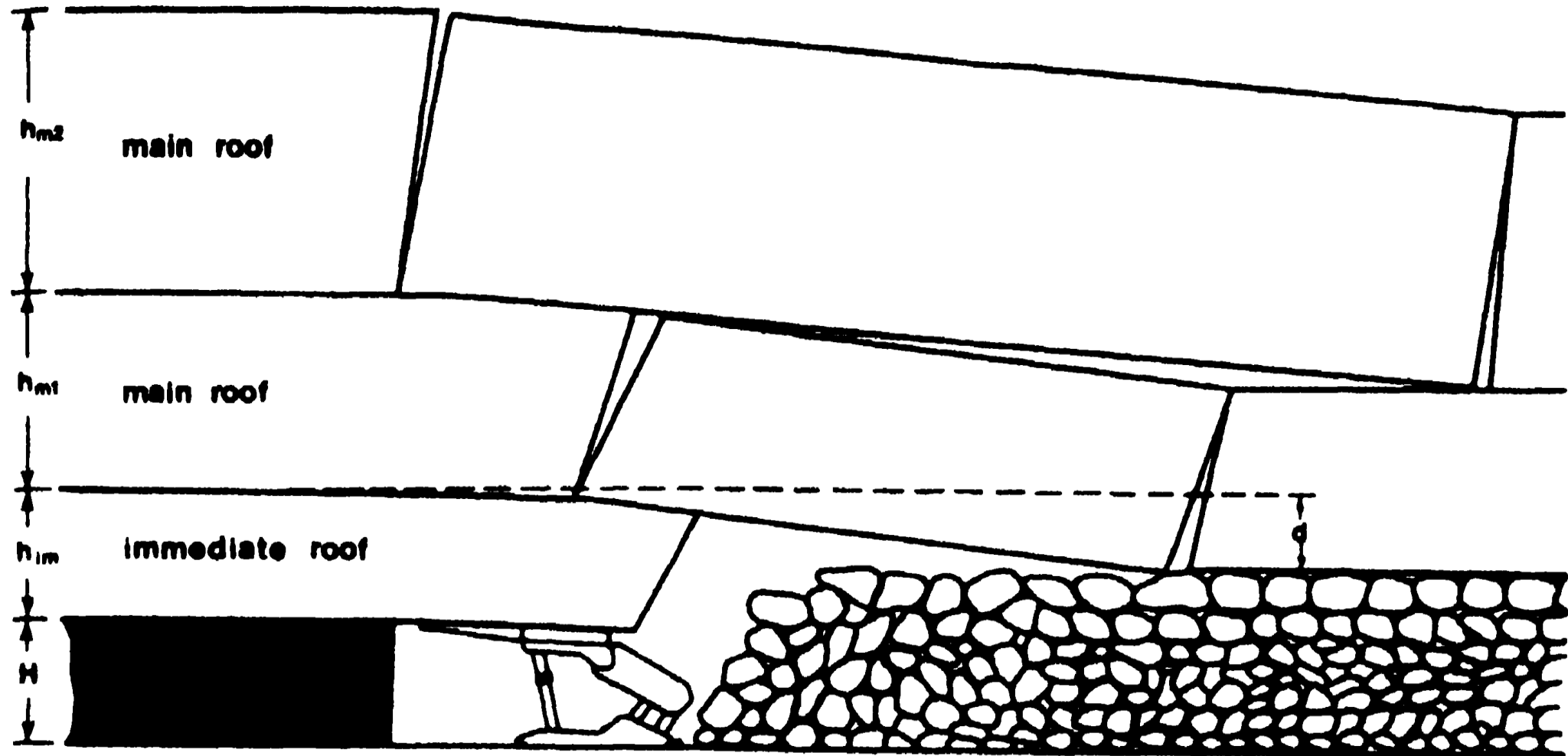
- ▶ Between the fractured zone and surface, is the **continuous deformation zone**
- ▶ In this zone, strata deform gently and without causing any major cracks that extend long enough to cut through the thickness of the strata, as in fractured zone
- ▶ The strata behaves like a continuous or intact medium
- ▶ On the surface, there is a **soil zone** of varying depth
- ▶ In this zone, the cracks are opened and closed as the longwall face comes and goes
- ▶ In general, cracks on and near panel edges tend to remain open permanently, whereas those around the centre of the panel will close back up when longwall face has passed by.



- ▶ Movement of the four zones has different degree of effect on roof control at longwall face
- ▶ The effect decreases as the strata located farther upward from roof line
- ▶ Those strata, movement of which will affect roof control, can be classified into **immediate roof** and **main roof**
- ▶ Immediate roof is that portion of overburden strata lying immediately above coal seam.
- ▶ Above immediate roof, lower portion of the fractured zone are called main roof
- ▶ Severity of impact on roof control at longwall face is proportional to their distance from the coal seam top or roof line.



# Immediate and Main Roof



# Caving height

- ▶ Thickness of immediate roof varies
- ▶ Caving initiates from the lowest strata in immediate roof and propagates upward into fractured zone
- ▶ Stratum sags downward as soon as it is mined out
- ▶ When downward sag exceeds the maximum allowable limit, it breaks and falls
- ▶ As it falls, its volume increases
- ▶ Gap between top of rock piles and sagged but uncaved stratum continues to decrease as the caving propagates upward
- ▶ When the gap vanishes, caving stops

$$H - d = h_{im} (K-1)$$

$$\text{And } d \leq d_0$$

Where  $H$  = mining height,  $d$  = sagging of lowest un-caved strata,  $D_0$  = maximum allowable sagging of lowest un-caved strata,  $H_{im}$  = thickness of immediate roof or caving height,  $K$  = bulking factor

$$H_{im} = (H - d) / (K-1)$$

- 
- ▶ Bulking Factor,  $K$ , is defined as the ratio of volume of broken rock strata to original volume before the same are broken and cave
  - ▶ Bulking factor is usually larger than one.
  - ▶ Bulking factor varies with rock type, shape and size of caved rock fragments, the ways the caved rocks piled up and the pressure imposed on the rock fragments.
  - ▶ When caving height is large and caved fragments are small, their arrangement is very disorderly, resulting in larger bulking factor.
  - ▶ Conversely, if caving height is small and caved fragments are large, their arrangement is very orderly and uniform, resulting in smaller bulking factor.



# First Weighting

- ▶ As longwall mining moves along the direction of mining, there are two distinct phases of overburden movement
- ▶ First phase of movement includes the distance from the setup entry to the point when the main roof first begins to break or if immediate roof does not cave right after support advance, final stage of an interval begins with a large area caving of immediate roof and lasts until the complete breakage of upper strata in main roof
- ▶ During such interval, maximum roof pressure measured at the face is called First Weighting
- ▶ The distance from the setup entry to first weighting is defined as First Weighting Interval,  $L_0$





# Periodic Weighting

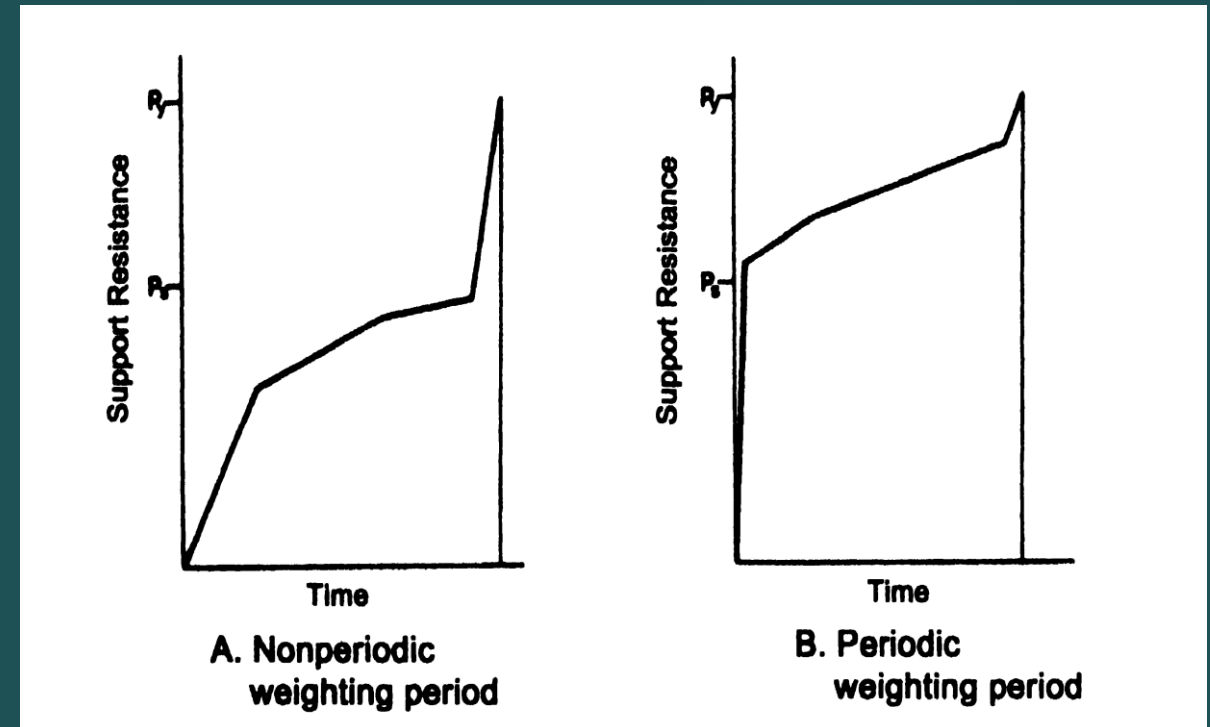
- ▶ The **second phase** begins **right after first weighting** and extends to the completion of the panel
- ▶ During this period, **roof pressure in the face area increases and decreases cyclically** due to **cyclical breakage of the immediate roof or main roof or both**
- ▶ This phenomenon is called **Periodic Weighting**
- ▶ Maximum pressure occurring in each period is **periodic roof weighting pressure**, and the distance between two consecutive roof weightings is called Periodic weighting interval,  $L_p$ .
- ▶ **Severity of first and periodic weighting depends very much on mining ad geological conditions**

# Idealized sequences of roof caving illustrating first caving and periodic caving



# Methods of identifying periodic roof weighting

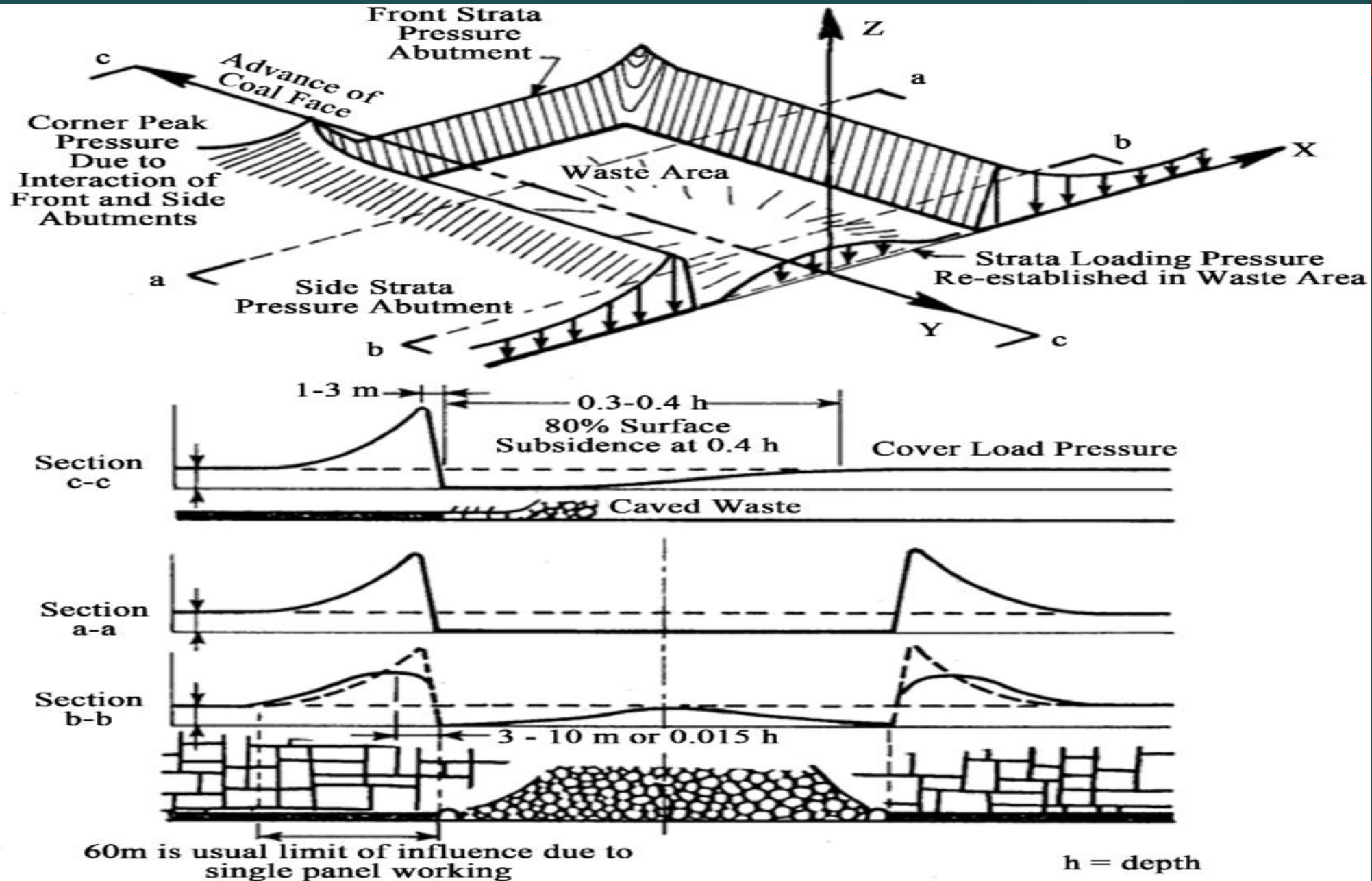
- ❑ Significant increase of time weighted average resistance (TWAR).
- ❑ Both maximum resistance and TWAR of a support cycle rise greatly during periodic weighting
  - ❑ Periodic weighting intensity  $I = P_{tm} / P_t$
  - ❑ Where  $P_t$  is the average TWAR during non-periodic weighting period immediately preceding the periodic weighting where  $P_{tm}$  is measured
- ❑ High Setting Resistance or Rapidly rising rate of resistance increase



# Abutment pressure

- ▶ Conceptually, a softer coal is sandwiched between relatively stronger roof and floor rocks and loaded by the weight of overburden
- ▶ Tress is uniformly distributed in the coal seam
- ▶ When panel gateroads are developed, the equilibrium conditions are destroyed due to presence of openings
- ▶ Stress distribution has to be readjusted to achieve a new equilibrium
- ▶ A de-stressed zone occurs in the roof of the entries and the load is transferred into neighboring solid coal both in panel and pillars
- ▶ The zones where vertical pressure exceeds the average overburden pressure before mining are created in or near the edges of the panel and pillars.
- ▶ These zones are called **abutments** and the above average pressures are the **abutment pressures**.

# Abutment pressure





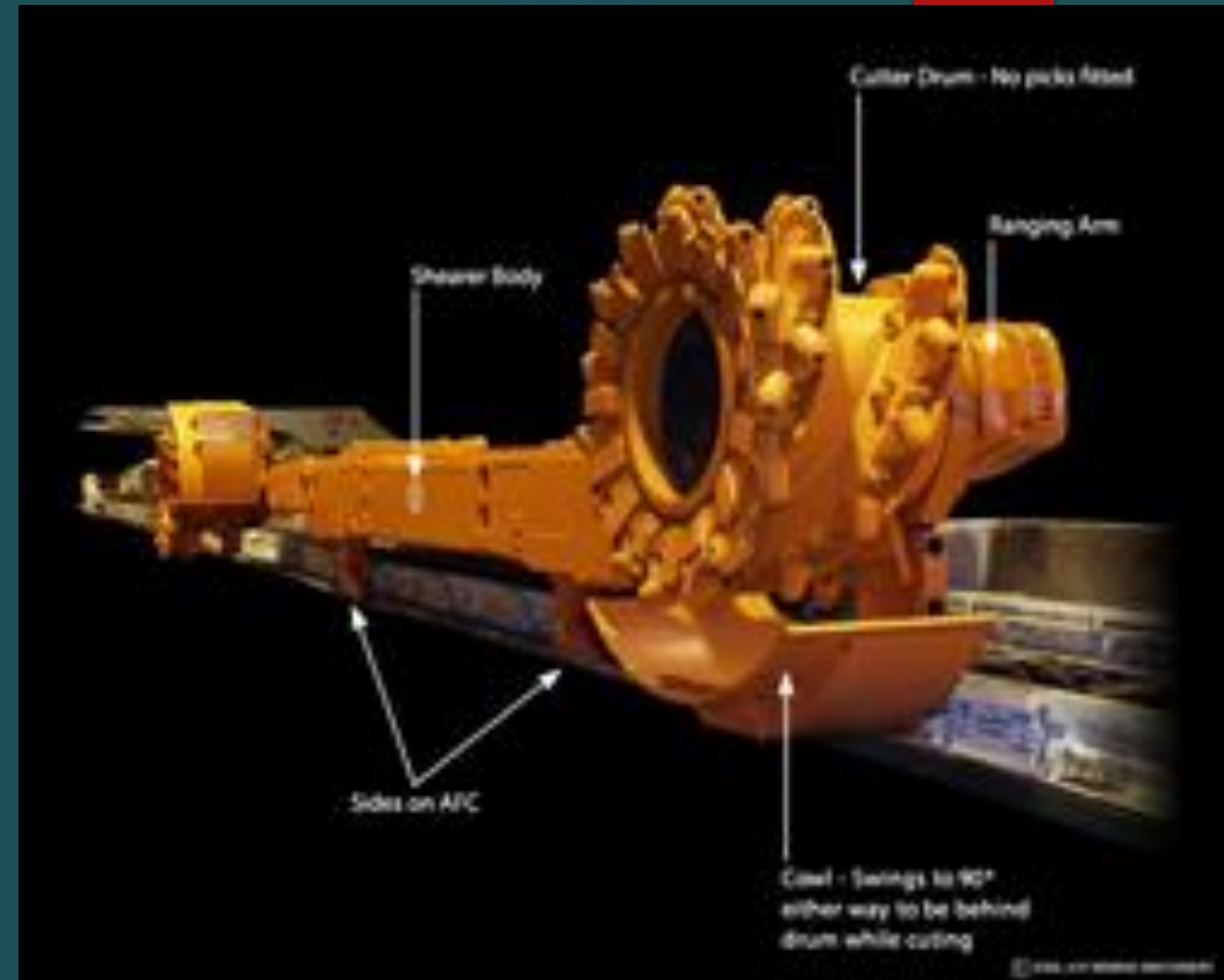
## Cutting Machines - Shearers

- ▶ A shearer consists of a machine body containing electric motors, hydraulic equipment and controls which is mounted over the AFC
- ▶ Horizontal cutting drums are mounted on the face side of the machine, laced with cutting picks and rotating in a plane parallel to the face
- ▶ If the AFC is pushed towards the face as the cutting drums are rotated and the shearer travels along the face, it is able to cut into the face for the full web width, moving along a snake in the AFC
- ▶ This is known as "**sumping in**"
- ▶ Once fully into the web, the shearer can advance the full length of the face cutting out the web.
- ▶ The snake can also be reversed to cut the wedge shaped portion of coal left while sumping in



# Cutting Machines - Shearers

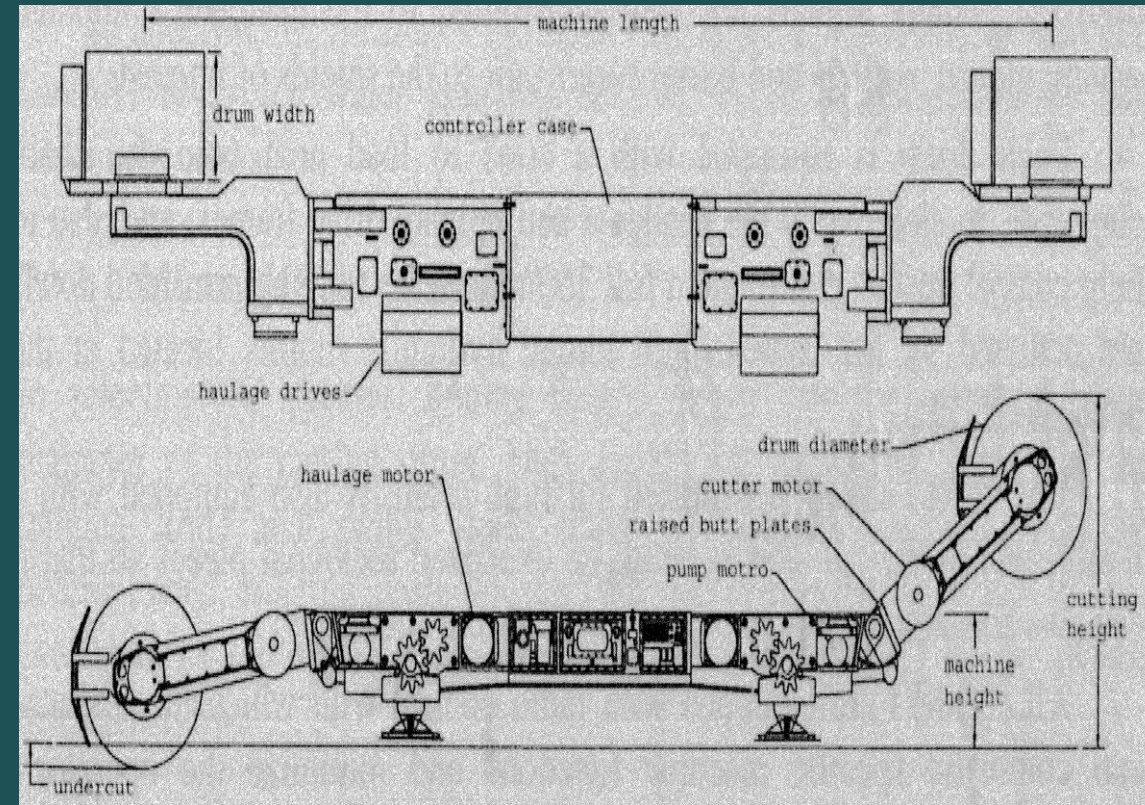
- ▶ The first shearers had a single drum rigidly attached to the body of the machine and so had a fixed cutting height and had to be able to get the body of the machine past the edge of the face at one end to cut the full length.
- ▶ Further developments saw the cutting drum being mounted on a hydraulically operated arm to vary the possible cutting height and a second cutting drum being fitted to provide a drum at both ends of the machine.
- ▶ Such machines were originally known as **"Double Ended Ranging Drum Shearers or DERDS"** but being almost the only type of shearer now used, the term "shearer" is generally understood to refer to this configuration.



# Cutting Machines - Shearers

## Shearer Haulage

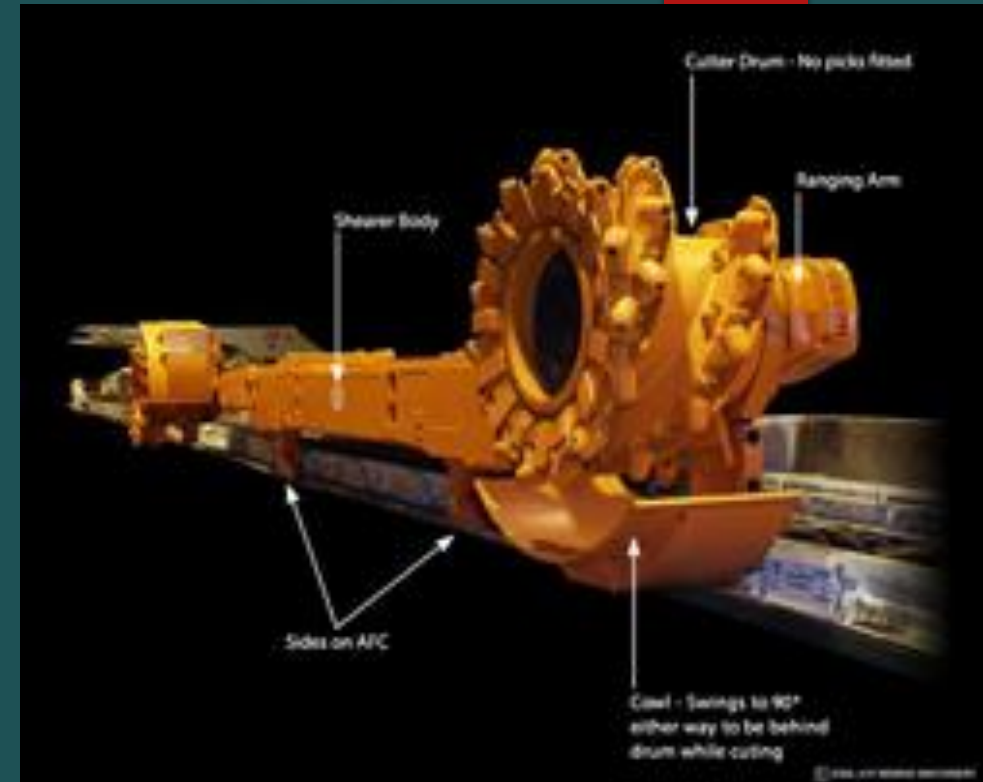
- ▶ Chain haulages have been used to drive shearers along the face
- ▶ Modern units use a rack and pinion arrangement for this purpose.
- ▶ Rack sections are attached on the face side of the AFC back plates and drive pinions, usually hydraulically driven, are attached to the goaf side of the shearer body.
- ▶ Stops are built into or placed in the rack sections at each end of the face to limit shearer travel and so prevent damage to other equipment in those locations
- ▶ Another device used to improve coal loading is a "**cowl**".
- ▶ This is a curved steel plate mounted on supporting arms so that it sits behind a cutting drum to prevent coal passing from the back of the drum into the area where the web has been cut, the coal being forced to move sideways towards the AFC instead.
- ▶ The cowl is fitted with a motor and drive pinion so that the cowl can be rotated from one side of the drum to the other when the cutting direction is reversed.





# Cutting Machines - Shearers

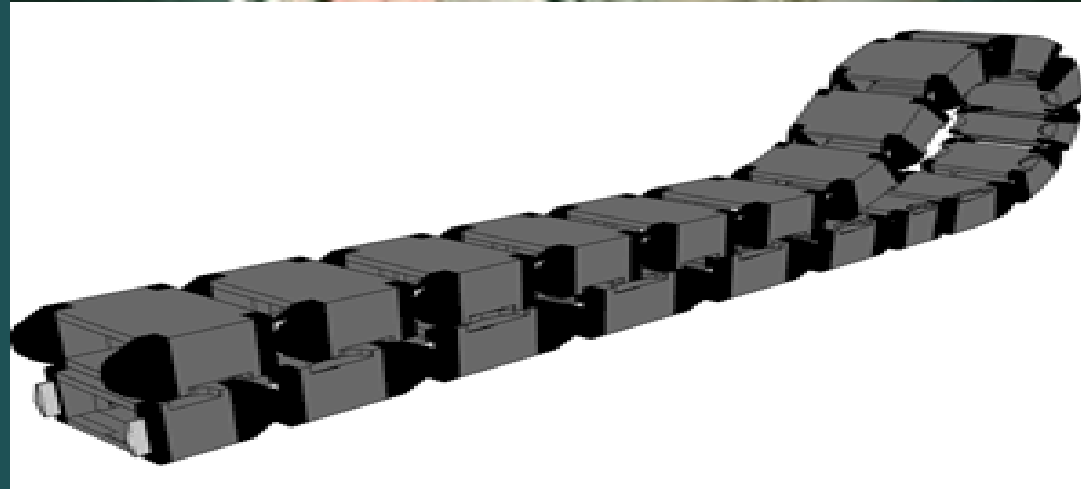
- ▶ Some shearer drums now have a hollow shaft venturi ventilation system design to ventilate the area on the face side of the drum while cutting, thereby preventing a possible ignition of gas given off in the cutting process.
- ▶ High pressure water is directed through the drums to the picks for dust suppression and pick cooling purposes, including water which has been used for cooling the shearer motors.
- ▶ Additional sprays are also fitted on the shearer for dust control, often acting as curtain sprays to keep dust on the face side of the shearer away from personnel rather than actually trying to settle the dust.



# Cutting Machines - Shearers

## Cable and Hose Handling (Bretby)

- A shearer has to be supplied with electric power and water and as it is continuously moving, trailing cables and hoses are required
- A shearer is continuously moving backwards and forwards along the entire face length and in a very congested area
- The "**Bretby**" is a flexible carrier consisting of a series of flat "plates" usually of a plastic material connected in the form of a chain but which only flexes along its length



# Plow for cutting coal

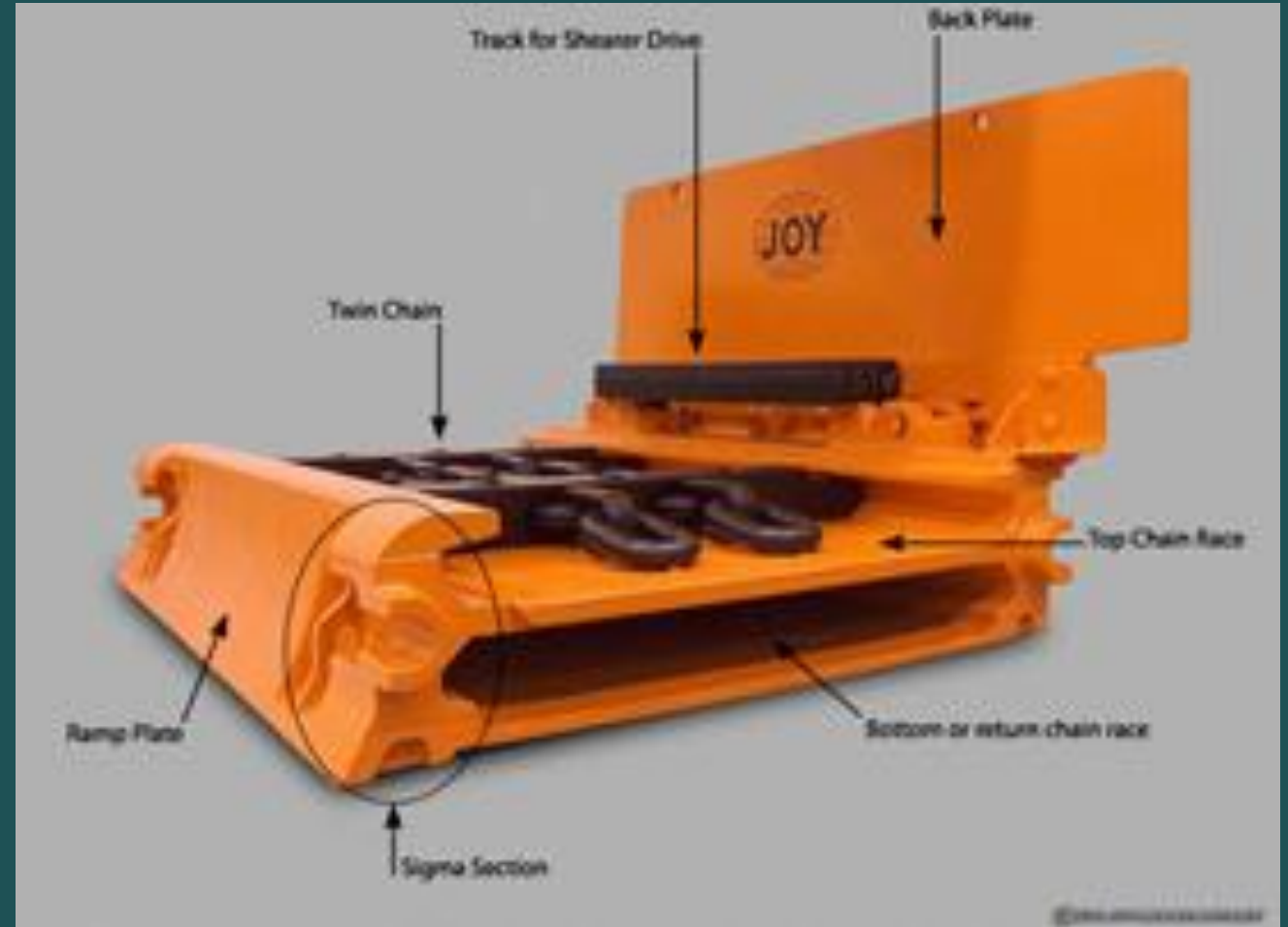
- ▶ For thin seams
- ▶ Coal is brittle and breaks in lumps
- ▶ Weak roof
- ▶ Hard or soft floor
- ▶ Steep seam can be worked





# Armoured Face Conveyor (Face Coal Haulage)

- ▶ AFC is a one-sided trough scraper conveyor,
- ▶ Cut coal falls into this trough which has an endless chain with scraper flights attached running along the base plate and returning below the base plate in an enclosed lower section or "**race**".
- ▶ The coal is dragged along the base plate by the flights.
- ▶ Early AFC's had two outboard chains, one running each side of the base plate with flights connecting them.
- ▶ Later developments saw the two outboard chains replaced by a single centre chain with flights cantilevered each side. With increasing longwall size, the single centre chain has now become a double or triple centre chain on almost all longwalls.





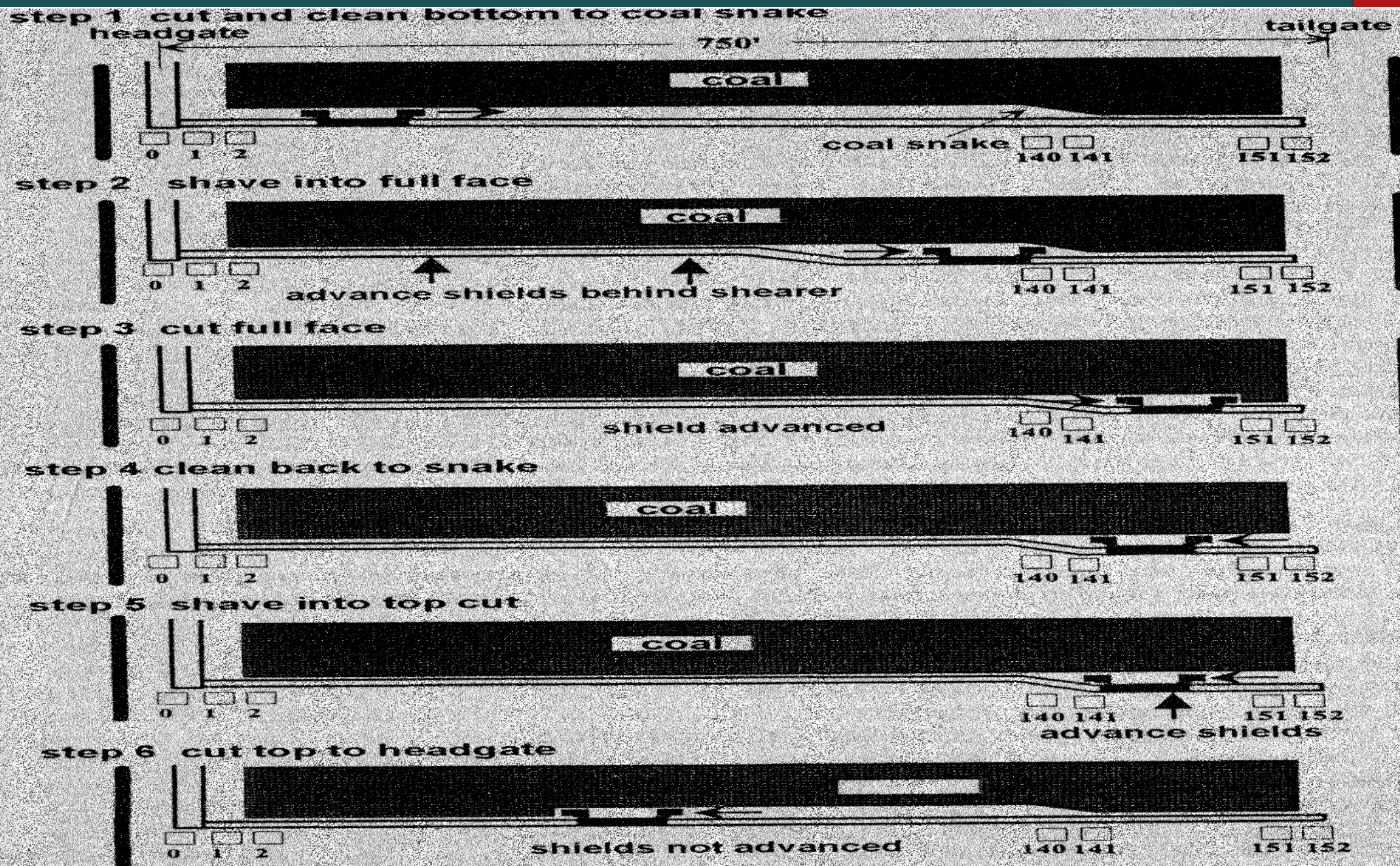
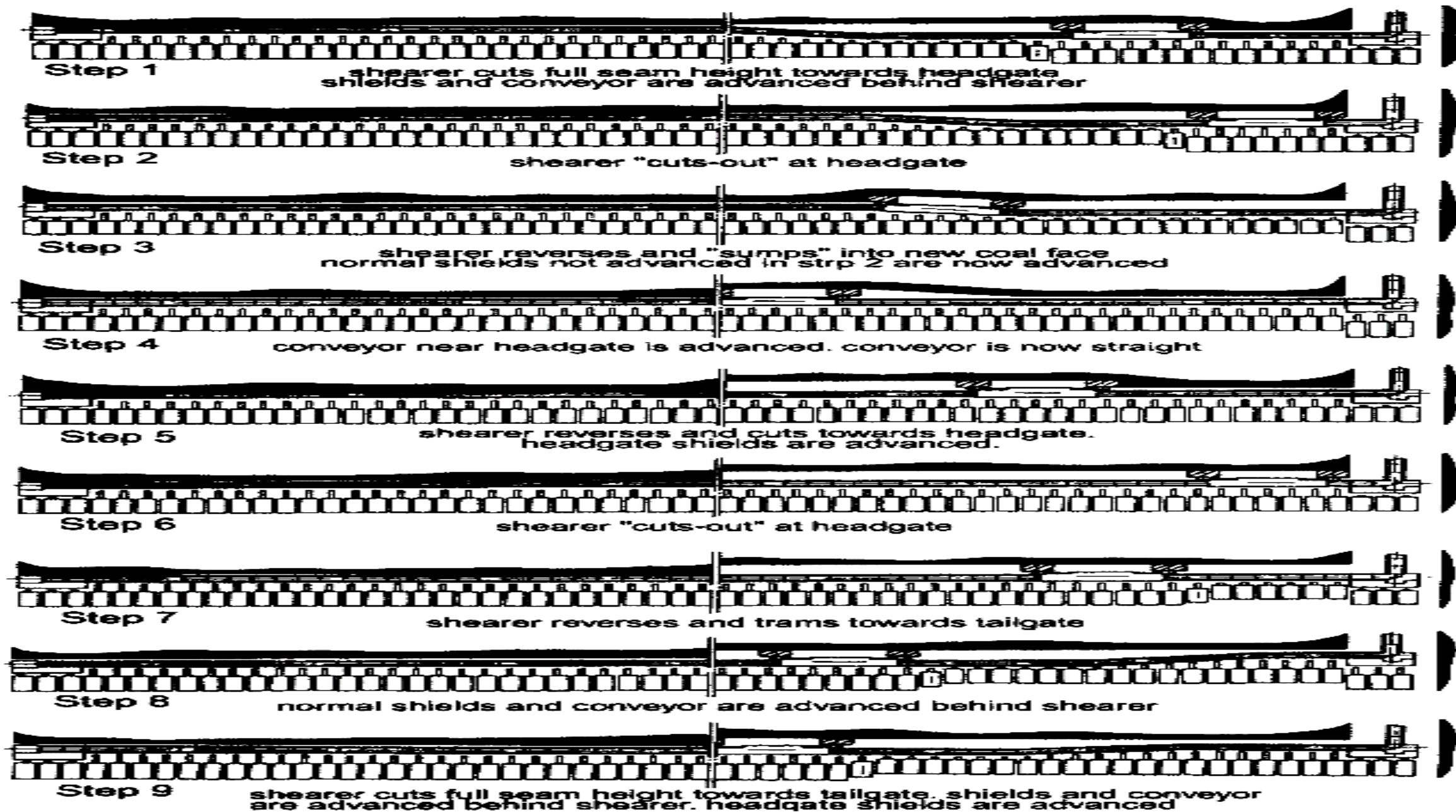


Figure 7.7.5 Modified uni-di cutting sequence



## B Full Face Cutting





Step 10

shearer approaches tailgate



Step 11

shearer "cuts-out" at tailgate



Step 12

shearer reverses and "sump" into new coal face  
normal shields not advanced in Step 11 are now advanced



Step 13

conveyor near tailgate is advanced  
conveyor is now straight



Step 14

shearer reverses and cuts towards tailgate  
tailgate shields are advanced



Step 15

shearer "cuts-out" at tailgate



Step 16

shearer reverses and trams towards headgate



Step 17

normal shields and conveyor are advanced behind shearer



Step 18

shearer cuts full seam height towards headgate, shields and  
conveyor are advanced behind shearer. tailgate shields are advanced

# Basic performance requirements for longwall

- ▶ Headgate and Tailgate
  - ▶ The headgate and tailgate must be kept straight, parallel and at the correct width
  - ▶ Failure to do so results in variation in width of longwall face resulting in gaps between support or congestion of supports in face
- ▶ Maintain straight line at face
  - ▶ Coalface, panline, and shields should form three parallel lines at face
  - ▶ If face is not straight, shields will walk into each other and panline is subject to bending and will wear more quickly
  - ▶ All pans must be pushed equally

# Basic performance requirements for longwall

- ▶ Keep the face area clean
  - ▶ To allow each piece of equipment to work to its capacity
  - ▶ Piling of debris in front of AFC and on shield base plates, which can obstruct shield advance and make shield move in different direction.
  - ▶ Desired distance of face advance in each cycle will not be achieved
  - ▶ Top of shield should be cleaned to reduce the loss of effectiveness of transmitting leg force to roof
- ▶ Cut the roof and floor smooth
  - ▶ Cutting horizon should be maintained smooth to avoid severe steps between webs and from shield to shield both in roof and floor
  - ▶ Shield canopy should make better contact with roof for better support and will cause less crushing

# Basic performance requirements for longwall

- ▶ Maintain setting pressure
  - ▶ Since setting pressure is the minimum pressure required to support the roof, it must be maintained at all times
  - ▶ Drop in setting pressure results into fall of roofs
- ▶ Keep the canopy tip to faceline distance minimum
  - ▶ For weak immediate roof, it is important to keep the distance from canopy tip to faceline as short as possible.
  - ▶ If it is more, roof falls take place in front of the supports

## Case Study - **LONGWALL MINING AT JHANJRA PROJECT**

- ▶ Lonwall mining started in this mine from November, 1989 in R-VII seam after procurement of two sets of powered support from Ex-USSR (KM-130) in 1989-90.
- ▶ Second set was commissioned in December, 1990.
- ▶ Another three sets were brought to Jhanjra mine from other collieries of ECL and South Eastern Coalfields Limited (SECL).
- ▶ All these sets worked successfully in R-VII/R-VIIA seam and finally salvaged to surface.



## Case Study - Jhanjra

- ▶ Geo-mining details of the Longwall Panel:
  - ▶ Length of Panel: 1652 m
  - ▶ Width of the Panel: 145 m (Centre to Centre)
  - ▶ Seam Thickness: 4.5 m to 5.5 m
  - ▶ Depth of Cover: 160 m to 170 m
  - ▶ Gradient of the seam: 1 in 16
  - ▶ Degree of gassiness: Degree-I
  - ▶ Presence of a fault: 3.5 m throw

# Details of Hydraulic Powered Support

| Sl. No. | Particulars                  | Remarks/ Dimensions       |
|---------|------------------------------|---------------------------|
| 1       | Support Type                 | 2-legged shield support   |
| 2       | Height of support            | 2600-5600 mm              |
| 3       | Support yield load @43.8 Mpa | 11000 KN                  |
| 4       | Setting Load @31.5 Mpa       | 7913 KN                   |
| 5       | Support center distance      | 1750 mm                   |
| 6       | Support Width range          | 1650-1850 mm              |
| 7       | Support density              | 1.34-1.37 Mpa (H=3.5-5.4) |
| 8       | Weight                       | 35 Tonne                  |

## Details of Shearer

| Sl. No. | Particulars              | Remarks/ Dimensions |
|---------|--------------------------|---------------------|
| 1       | Production capacity      | 2800 Tonne/hour     |
| 2       | Diameter of Shearer drum | 2800 mm             |
| 3       | Maximum cutting height   | 5500 mm             |
| 4       | Lump coal size           | 1129 mm             |
| 5       | Drum web                 | 800 mm              |
| 6       | Haulage cutting speed    | 0-17.1 m/ min       |
| 7       | Rotation speed of drum   | 29.76 r/ min        |
| 8       | Weight                   | 95000 kg            |

# Details of Road header

| Sl. No. | Particulars            | Dimensions  |
|---------|------------------------|-------------|
| 1       | Maximum Cutting Height | 4.8 m       |
| 2       | Maximum Cutting Width  | 5.0 m       |
| 3       | Climbing capacity      | +/- 18 deg. |
| 4       | Speed of travel        | 0-8 m/ min  |
| 5       | Rated Voltage          | 1100V       |
| 6       | Total Power            | 286 KW      |
| 7       | Weight                 | 45 Tonne    |

# Transport & Installation of Longwall equipment using Diesel Vehicles

- ▶ Transportation of entire longwall equipment from surface to underground was done within a span of 23 days
- ▶ Entire machinery was installed within a month.
- ▶ Multi Utility Vehicle (MUV) were used for the purpose of quick and safe transport of equipment.
- ▶ To facilitate transportation of Longwall equipment using diesel vehicles, Incline-2 (total length-1070 m up to R-V seam) was concreted with M-40 grade self-compacting concrete.

# Transport & Installation of Longwall equipment using Diesel Vehicles

- ▶ Three types of MUV (CL210, CL215 and SH660D) manufactured and supplied by M/S Caterpillar are being used at Jhanjra Project for transportation of material and machinery.
- ▶ CL210- This machine can be attached with Bucket, Fork, Trailer which can carry a load of 20 tonnes and a basket with a maximum capacity of 1 tonne.
- ▶ CL215-This machine can be attached with Bucket, Cookie Plate which can carry a load of 20 tonne and a Trailer with maximum capacity of 40 tonnes.
- ▶ This is specifically used for transportation of Powered Supports of 37 tonnes each.





# Transport & Installation of Longwall equipment using Diesel Vehicles

- ▶ SH660D-This machine can carry maximum load of 60 tonnes with fork attached at the front
- ▶ This is mainly used for squaring of Powered Supports in correct position after unloading.



# Transport & Installation of Longwall equipment using Diesel Vehicles

- ▶ Chock Shield/ Roof Support Trailer (SH150) –
- ▶ These were used for transport of Chock Shields with CL215 at Jhanjra Project.
- ▶ Support mass of SH150 is 55 tonnes.
- ▶ The trailer has a hydraulic powered winch for loading the Chock/Roof Support into the trailer body.
- ▶ The floor of the trailer is tapered allowing easy access into the trailer body by the support.



# Transport of men by Drift Runner

- ▶ Transport of men by Drift Runner (Model-G series) –
- ▶ For the purpose of transport of men from pit bottom of Shaft-A, three Drift Runners (1 for Longwall panel and other 2 for CM panels) are being used in this project.
- ▶ These underground men transporters have been supplied by VLI Diesel Pty Limited, Australia.
- ▶ This can accommodate 12 persons at a time.



# Production from Longwall Panel

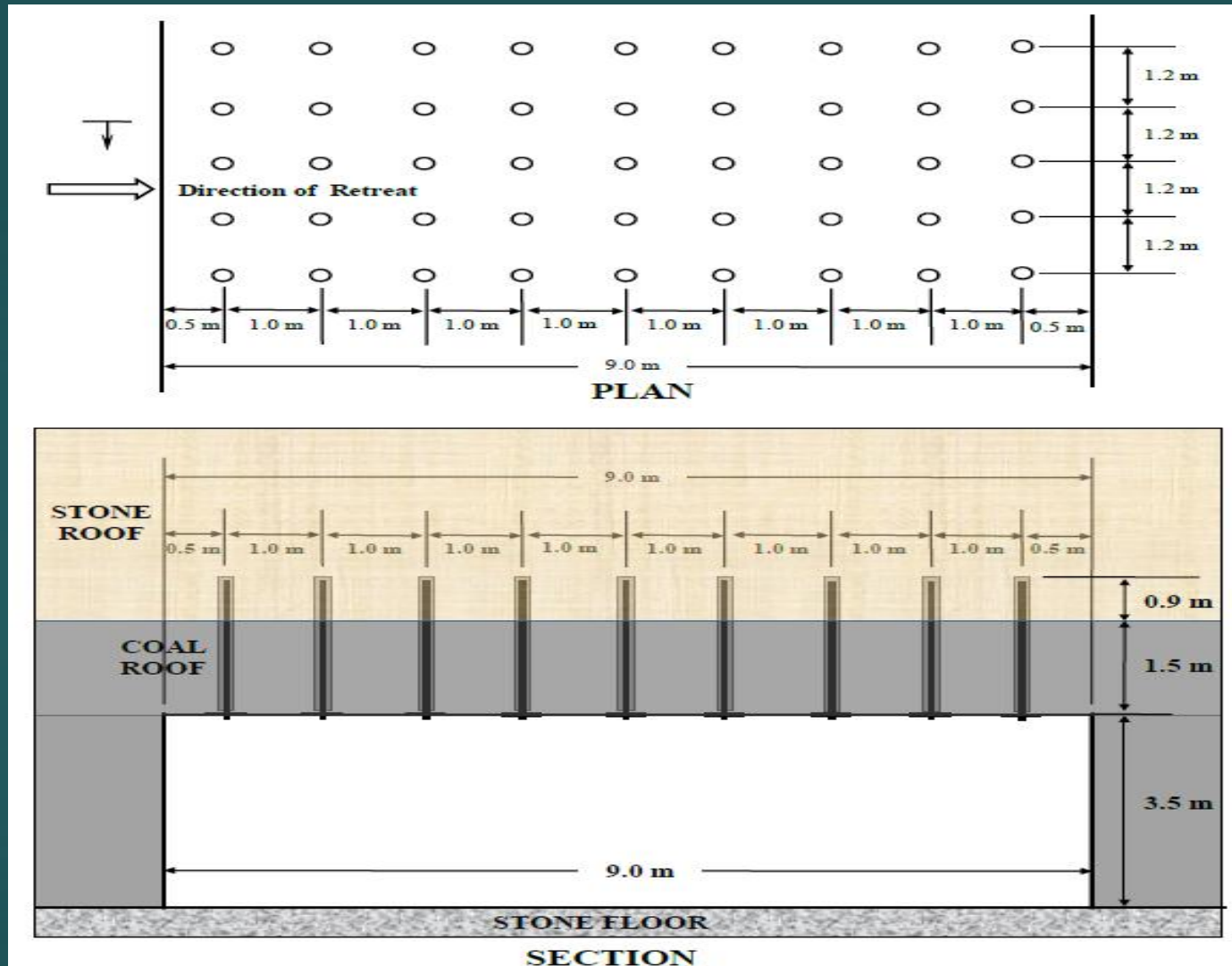
- ▶ During 2016-17, (from August, 2016 to March, 2017) production from LW panel was 0.95 MT.
- ▶ Average daily production from the LW panel is 6000-6500 TPD.
- ▶ In addition, a production of about 500-550 TPD is being obtained from gate roads development by road headers.
- ▶ A maximum production of 8360 TPD has been achieved from Longwall panel.

# Gate Roads

- ▶ Main Gate: 3.8 m (height) x 4.8 m (width)
- ▶ Tail Gate: 3.6 m (height) x 4.8 m (width)
- ▶ The gate roadways have been supported with two rows of 40T capacity hydraulic props at 1.5 m interval up to a distance of 30 m from the LW face in addition to the supports erected during development.



# Schematic diagram for installation chamber with its pattern of roof bolt supports





# Salvaging Chamber

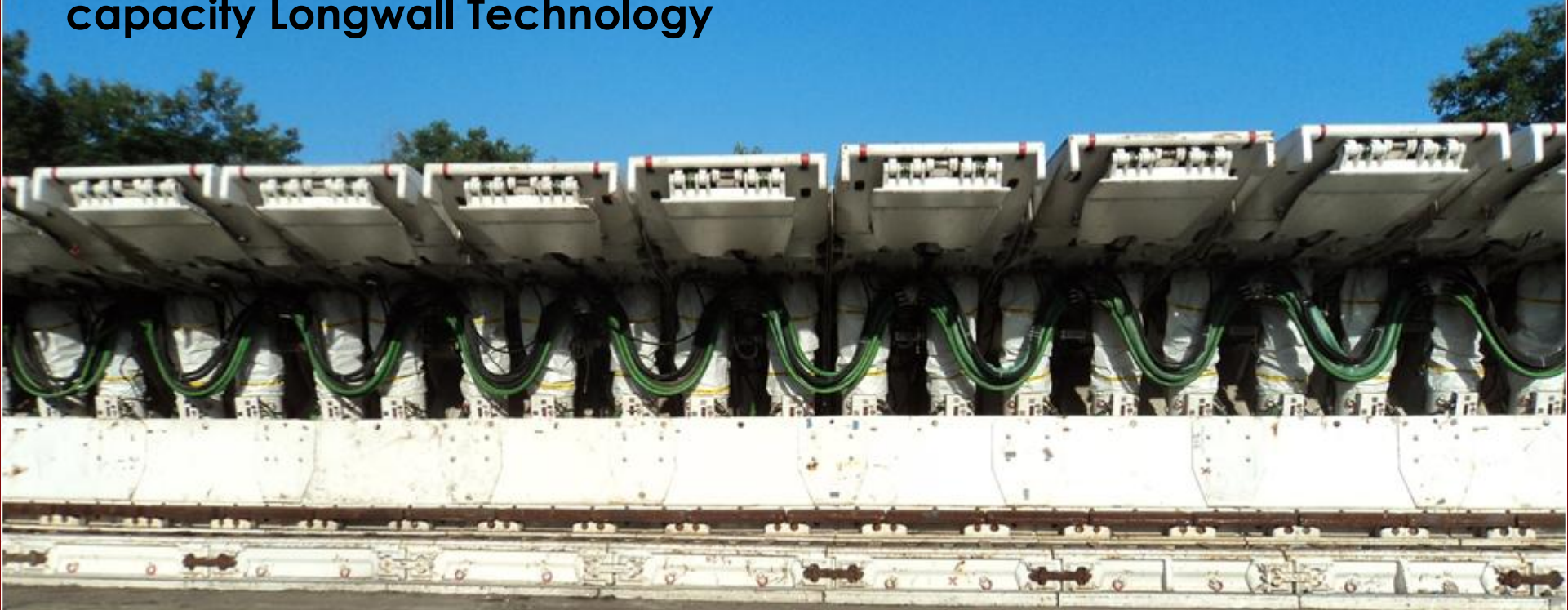
The salvage chamber was prepared by reducing the working thickness to 3.2 m with support of roof bolts in a grid pattern of 0.8 m x 1.75 m

W-straps were incorporated with the roof bolts for the free face length of last 6 m of the salvage chamber out of the total 9 m width (of the salvage chamber).

Timber cogs was set at the goaf edge after withdrawal of powered support to take protection against influx of debris at the site of support withdrawal wherever necessary.

# Promising a brighter future

SCCL Enters into a new orbit of High capacity Longwall Technology



The Singareni Collieries  
Company Ltd.,  
(A Government Company)



# INTRODUCTION

## A L P



- SCCL produced about 61.33 Mt of coal during 2016-17 amount to about 9% of the Indian coal production.
- Against national average of about 9%, the Underground coal production in SCCL is about 15.51%.
- SCCL has more coal reserves at below 300m depth than up to 300m depth.
- The Longwall technology is the only technology for achieving bulk production from such conditions.
- Hence, Adriyala Project has attained such high significance.





## Adriyala Longwall Project

To meet the increasing demand from power sector and shrinking of ongoing opencast projects, SCCL envisaged the opening up of deep shaft blocks for large-scale production. At present the existing pithead 2600 MW power station of NTPC, Ramagundam is supplied by 3 OC projects of SCCL at Ramagundam (10.20MT) out of its total requirement of 12.50 MT. The reserves in these OC mines are depleting and opening of this 2.817MT capacity Adriyala Longwall project will help to sustain the coal supply from Ramagundam mines to retain the pithead NTPC Ramagundam Super Thermal Power Station.

■ The project commissioned on 15.10.2014.



A  
L  
P





# ALP



## Significance

- Coal Production in India is anticipated to grow rapidly as the country has shortage of other sources of energy.
- The present operating Under Ground and Opencast mines are exploiting coal reserves up to 300m depth and will be depleted in next 10-15years.
- Future mining has to be done beyond 300m depth with Under Ground methods.

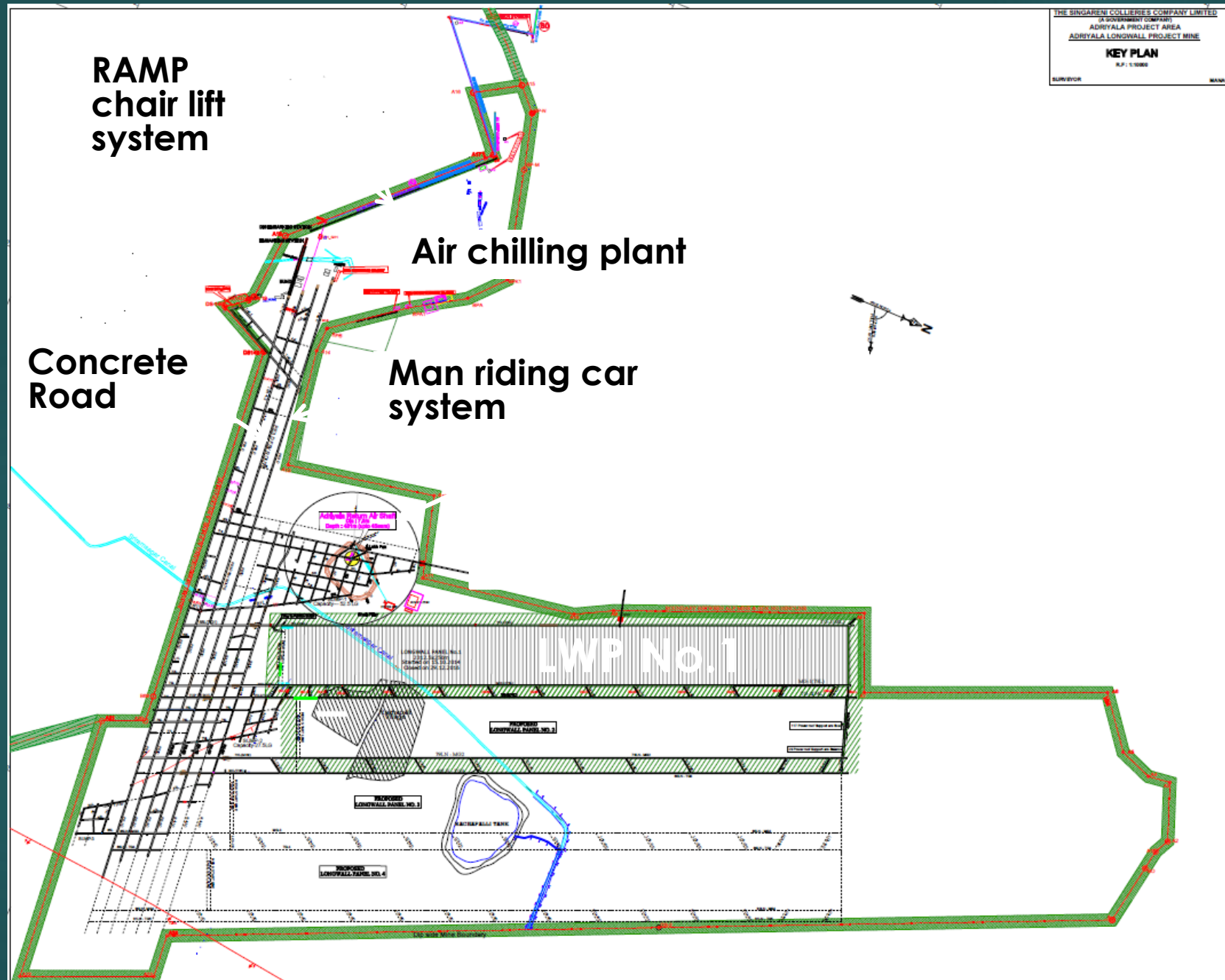
### Challenges in exploitation of deep deposits :

- High strip ratio (OC Mines)
- Greater strata control problems
- High investments
- Ventilation problems

Conventional Underground methods are not adoptable at greater depths –

**Longwall technology is a proven UG mining method for deep seated deposits.**

***Hence sustainable Under ground coal mining is possible only by successful adoption of Longwall mining.***







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# Salient features of Adriyala Longwall Project

|                                     |                      |
|-------------------------------------|----------------------|
| Total extractable Reserves          | : 78.597 MT          |
| Total Mine take area                | : 4.845 Sq. Km.      |
| Seam gradient                       | : 1 in 6 to 1 in 7.8 |
| Gassiness of the Mine               | : Degree-I           |
| Depth of working                    | : 294.0m– 644.0m     |
| Targeted production                 | : 2.817MTPA          |
| Life of the Mine                    | : 35 years           |
| Productivity<br>at Mine level (OMS) | : 12.35 T/Man shift  |



# Specifications of Equipment

## POWER ROOF SUPPORTS

- Support capacity : 1152 T (2 legged DTDA version)
- Operating Range : 2.8 to 3.6 m
- Support width : 1.75 m

## SHEARER

- Type / Model : EL 3000
- Operating voltage : 3.3 KV
- Web depth : 0.85 m
- Drum diameter : 2.3 m
- Total installed power : 2245 KW

## AFC

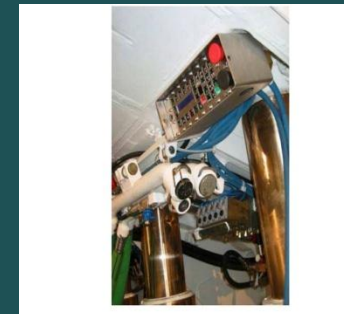
- Capacity : 3X855 KW, 3.3 KV, 3-Phase, 50 Cycles
- Width : 1142 mm
- capacity : 3100 TPH

## BSL

- Capacity : 400KW, 3.3 KV, 3-Phase, 50Cycles/ sec.
- Width : 1342mm wide,
- speed : 2.06 m/sec
- capacity : 3200 TPH.

## Electricals

- 2 X 4.5MVA transwitches



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## Outbye conveyor system - CODCO

- Total number of : 11 nos
- Gear heads
- Capacities : 1 X 315KW to  
varying from 4 X 500KW
- Speed : 4 m / sec
- Type of Drive : VFD
- Belt capacity : 3500 TPH
- Belt Type : Steel cord (ST 800 to ST2500)
- Total Length : 9km
- Total installed power : 12765 KW



## Equipment suppliers - ALP

Order placed on M/s Caterpillar in supply of  
Complete Longwall equipment package  
vide order No: 7700000111 dt: 18.12.2009.

- LW face (main) equipment
- Gate belt
- 11 KV Electricals, Cables for 11 KV system
- Diesel Vehicles & Spares for transport  
equipment

### Other Major equipment

- Road headers
- 120T tyre mounted Crane
- Ventilation Fans
- Over land and Trunk conveyors
- Cable bolts and its accessories
- Panline Bolter



: M/s Caterpillar, Germany

: M/s Caterpillar, USA

: M/s B&F, UK

: M/s Caterpillar, Australia

: M/s DOSCO overseas, UK

: Groove, US

: M/s Zitron, Spain

: M/s CODCO, China

: M/s DSI, Australia

: M/s Sandvik, Austria



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## New & Challenging activities carried out at Adriyala

### Development of roadways with bigger dimensions:

- A total drivage of about 52 kilometers is done.
- The dimensions of the roadways are larger - (5.2 to 5.5 m width and 3.2 to 3.6 m height) than conventional. The support systems are improved to suit this.
- The installation face is widened to 8 m and could be successfully supported by adopting cable bolts for the first time.

### Outbye conveyor system:

- High speed belt conveyors were installed for a length of 9.0 km from the long wall face with a total number of 11 no's Gear heads with about 12 MW power.
- Huge civil foundations are prepared some of which are in loose OB soil are done for their installation.



# New & Challenging activities carried out at Adriyala



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## Ventilation

- Ventilation is provided by an underground fan during development. A deep shaft of about 470 m is sunk under adverse conditions.
- Two main fans of 400kw and 15,000 Cu.m/min and four booster fans of 75kw of 4,000 Cu.m/min capacities were procured from M/s Zitron, Spain and installed.
- An outsourced air chilling plant of 1400TR was installed at mouth of Punch Entry-5 (PE-5). About 3500Cu.m/min of chilled air is being supplied at 90C at the entry of PE-5.

## 11KV Electrical System

- 11 kV transmission to UG - **First time in India.**
- VFD drives and Auto power factor correction unit is installed for the first time which is designed to curtail the wasteful energy consumption.





# New & Challenging activities carried out at Adriyala

## Floor Concreting

- For transport of Longwall equipment using diesel vehicles, one of the trunk road ways with weak floor was concreted with M-40 grade self compacting concrete for a length of about 2.0Km.

## Transport & Installation of Longwall equipment

- Total equipment transportation and installation was done by diesel vehicles from surface to Longwall face
- Approx. 9000 T of equipment was transported by these vehicles.

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## Uniqueness of ALP

The following new technologies have been introduced for the first time in SCCL:

- Punch entry (Direct access to coal from Opencast highwall)
- 11 KV power transmission to UG (for less voltage drop)
- Pre tensioned Cable bolting (For effective roof support)
- 400 KW high capacity fan commissioned (To supply more air with high water gauge)
- Floor Concreting in underground



ALP





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## Uniqueness of ALP

The following new technologies have been introduced for the first time in SCCL:

- Automation systems (For sequence control/less manual intervention)
- Diesel transport vehicles (FBL) (For speedy & safe equipment transportation)
- VFD controlled un-manned Belt conveyor system (For soft start and power conservation)
- Mine cruiser( for faster transport of men)
- Surface air chilling (To create comfortable environment conditions)
- Bolter Miner (For faster rate of development)



## Details of longwall panel No.1

- Dimensions of panel : 2333m X 250m
- Depth range : 362m to 450m
- Reserves in the panel : 3.36 MT
- Life of the panel : 12 months
- Date of commissioning : 15.10.2014
- Highest daily production : 13187 T (29.2.16)
- Highest monthly production : 2.53 LT (Jan,2016)
- Avg. production : 8000-9000TPD

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# Adriyala Longwall Project The Singareni Collieries Company Ltd.,

